DE1-SoC Computer System with Nios® II

For Quartus® Prime 23.1

1 Introduction

This document describes a computer system that can be implemented on the DE1-SoC development and education board, which is described in the `Teaching and Projects Boards` section of the FPGAcademy.org website. This system, called the *DE1-SoC Computer with Nios II*, is intended for use in experiments on computer organization and embedded systems.

To support such experiments, the system contains embedded processors, memory, audio and video devices, and some simple I/O peripherals. The FPGA programming file that implements this system, as well as its design source files, can be obtained from the FPGAcademy's Design Example GitHub repository.

2 DE1-SoC Computer Contents

A block diagram of the *DE1-SoC Computer with Nios II* is shown in Figure 1. As indicated in the figure, the components in this system are implemented utilizing both the FPGA and the *Hard Processor System* (HPS) inside the Cyclone® V SoC chip. The FPGA implements two Nios II® processors and several peripheral ports: memory, timer modules, audio-in/out, video-in/out, PS/2, analog-to-digital, infrared receive/transmit, and parallel ports connected to switches and lights. The HPS comprises an ARM* Cortex* A9 dual-core processor and a set of peripheral devices. Some of these HPS peripheral devices can be accessed by Nios II. Instructions for using the HPS with the ARM processor can be found in the document entitled *DE1-SoC Computer System with ARM* Cortex* A9*, which is available on the FPGAcademy.org website.

2.1 Nios® II Processor

The Nios II processor is a 32-bit CPU that can be implemented in an Altera FPGA device. Two versions of the Nios II processor are available, designated economy (/e) and fast (/f). The DE1-SoC Computer includes two instances of the Nios II/f version, configured with floating-point hardware support.

An overview of the Nios II processor can be found in the document *Introduction to the Nios II Processor*, which is available in the `Computer Organization System Design` section of the FPGAcademy.org website. An easy way to begin working with the DE1-SoC Computer and the Nios II processor is to make use of a utility called the *Monitor Program*. It provides an easy way to assemble/compile Nios II programs written in either assembly language or the C language. The Monitor Program, which can be downloaded from the FPGAcademy.org website, is an application program that runs on the host computer connected to the DE1-SoC board. The Monitor Program can be used to control the execution of code on Nios II, list (and edit) the contents of processor registers, display/edit the contents of memory on the DE1-SoC board, and similar operations. The Monitor Program includes the DE1-SoC Computer as a predesigned system that can be downloaded onto the DE1-SoC board, as well as several sample

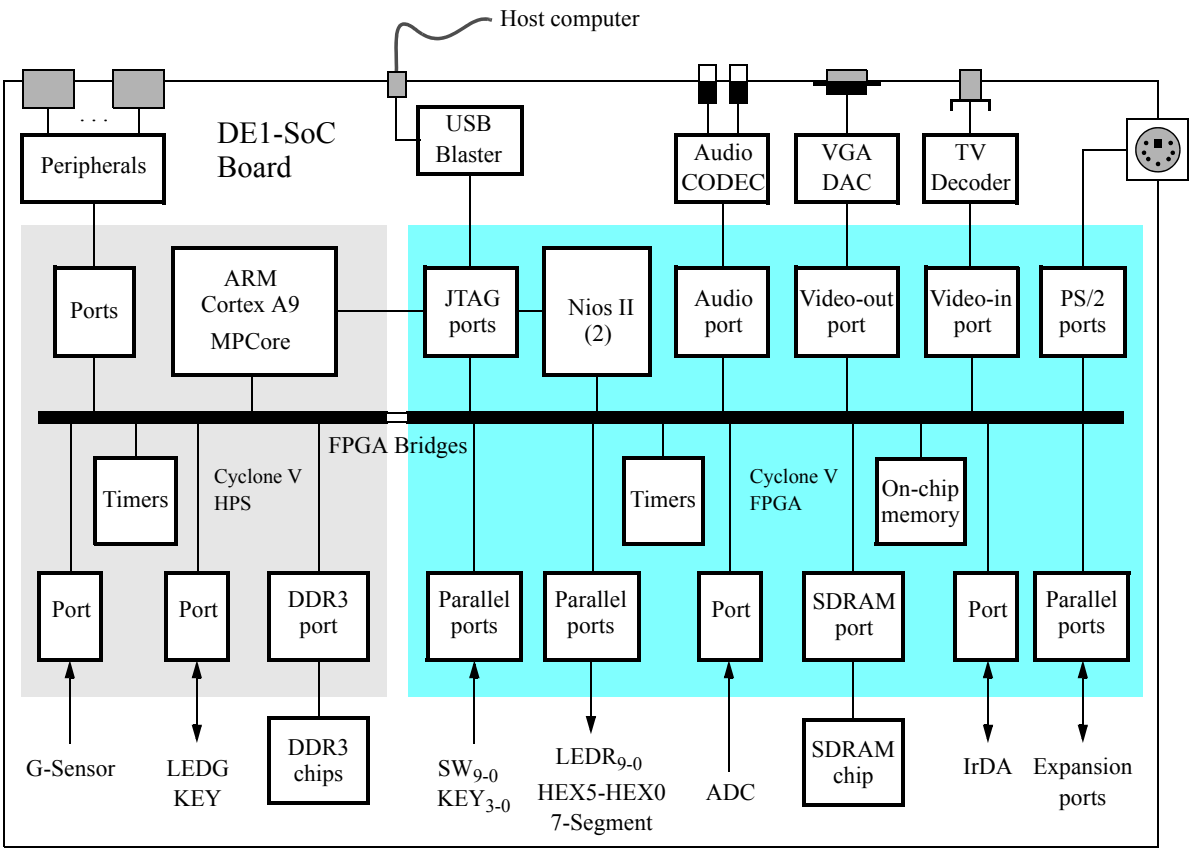


Figure 1. Block diagram of the *DE1-SoC Computer with Nios II*.

programs in assembly language and C that show how to use the DE1-SoC Computer's peripherals. Some images that show how the DE1-SoC Computer is integrated with the Monitor Program are described in Section 8.

All of the I/O peripherals in the DE1-SoC Computer are accessible by the processor as memory mapped devices, using the address ranges that are given in the following subsections.

2.2 Memory Components

The DE1-SoC Computer has DDR3 and SDRAM ports, as well as two memory modules implemented using the on-chip memory inside the FPGA. These memories are described below.

2.2.1 SDRAM

An SDRAM Controller in the FPGA provides an interface to the 64 MB synchronous dynamic RAM (SDRAM) on the DE1-SoC board, which is organized as 32M x 16 bits. It is accessible by the Nios II processor using word (32-bit), halfword (16-bit), or byte operations, and is mapped to the address space `0x00000000` to `0x03FFFFFF`.

2.2.2 DDR3 Memory

A 1 GB DDR3 memory is connected to the HPS part of the Cyclone® V SoC chip. The memory is organized as 256M x 32-bits, and is accessible using word accesses (32 bits), halfwords, and bytes. The Nios II processor can access the DDR3 memory using the addresses space 0x40000000 to 0x7FFFFFFF.

2.2.3 On-Chip Memory

A 256 KB memory is implemented inside the FPGA, organized as 64K x 32 bits. The Nios II processor can access this memory using addresses in the range 0x08000000 to 0x0803FFFF. This memory is used as a pixel buffer for the video-out and video-in ports.

2.2.4 On-Chip Memory Character Buffer

An 8 KB memory is implemented inside the FPGA for use as a character buffer for the video-out port, which is described in Section 4.2. The character buffer memory is organized as 8K x 8 bits, and spans the Nios II address range 0x09000000 to 0x09001FFF.

2.3 Parallel Ports

There are several parallel ports implemented in the FPGA that support input, output, and bidirectional transfers of data between the Nios II processor and I/O peripherals. As illustrated in Figure 2, each parallel port is assigned a *Base* address and contains up to four 32-bit registers. Ports that have output capability include a writable *Data* register, and ports with input capability have a readable *Data* register. Bidirectional parallel ports also include a *Direction* register that has the same bit-width as the *Data* register. Each bit in the *Data* register can be configured as an input by setting the corresponding bit in the *Direction* register to 0, or as an output by setting this bit position to 1. The *Direction* register is assigned the address *Base* + 4.

Address	31 30 . . . 4 3 2 1 0	
Base	Input or output data bits	Data register
Base + 4	Direction bits	Direction register
Base + 8	Mask bits	Interruptmask register
Base + C	Edge bits	Edgecapture register

Figure 2. Parallel port registers in the *DE1-SoC Computer with Nios II*.

Some of the parallel ports have registers at addresses *Base* + 8 and *Base* + C, as indicated in Figure 2. These registers are discussed in Section 3.

2.3.1 Red LED Parallel Port

The red lights $LEDR_{9-0}$ on the DE1-SoC board are driven by an output parallel port, as illustrated in Figure 3. The port contains a 10-bit *Data* register, which has the address 0xFF200000. This register can be written or read by the processor using word accesses, and the upper bits not used in the registers are ignored.

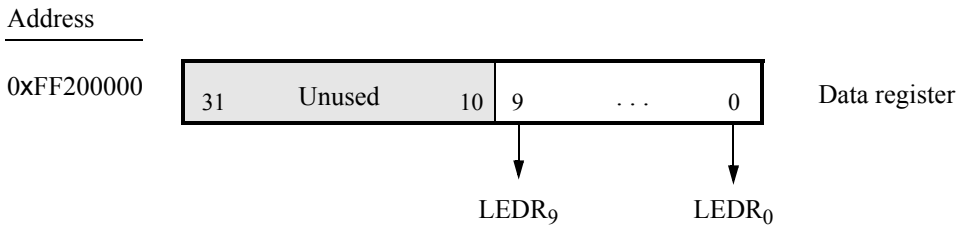


Figure 3. Output parallel port for *LEDR*.

2.3.2 7-Segment Displays Parallel Port

There are two parallel ports connected to the 7-segment displays on the DE1-SoC board, each of which comprises a 32-bit write-only *Data* register. As indicated in Figure 4, the register at address 0xFF200020 drives digits *HEX3* to *HEX0*, and the register at address 0xFF200030 drives digits *HEX5* and *HEX4*. Data can be written into these two registers, and read back, by using word operations. This data directly controls the segments of each display, according to the bit locations given in Figure 4. The locations of segments 6 to 0 in each seven-segment display on the DE1-SoC board is illustrated on the right side of the figure.

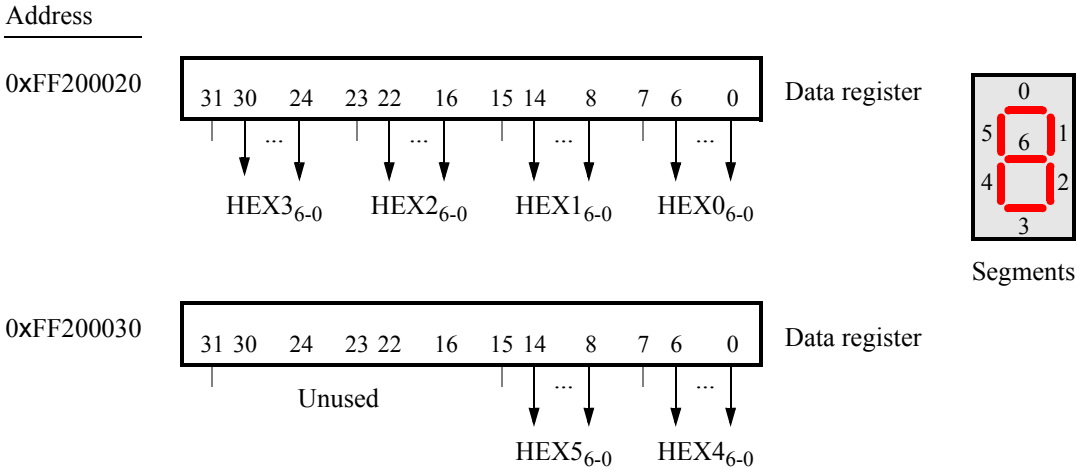


Figure 4. Bit locations for the 7-segment displays parallel ports.

2.3.3 Slider Switch Parallel Port

The SW_{9-0} slider switches on the DE1-SoC board are connected to an input parallel port. As illustrated in Figure 5, this port comprises a 10-bit read-only *Data* register, which is mapped to address 0xFF200040.

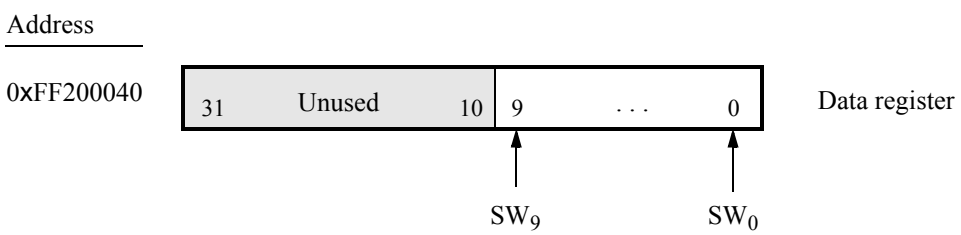


Figure 5. *Data* register in the slider switch parallel port.

2.3.4 Pushbutton Key Parallel Port

The parallel port connected to the KEY_{3-0} pushbutton switches on the DE1-SoC board comprises three 4-bit registers, as shown in Figure 6. These registers have the base address 0xFF200050 and can be accessed using word operations. The read-only *Data* register provides the values of the switches KEY_{3-0}. The other two registers shown in Figure 6, at addresses 0xFF200058 and 0xFF20005C, are discussed in Section 3.

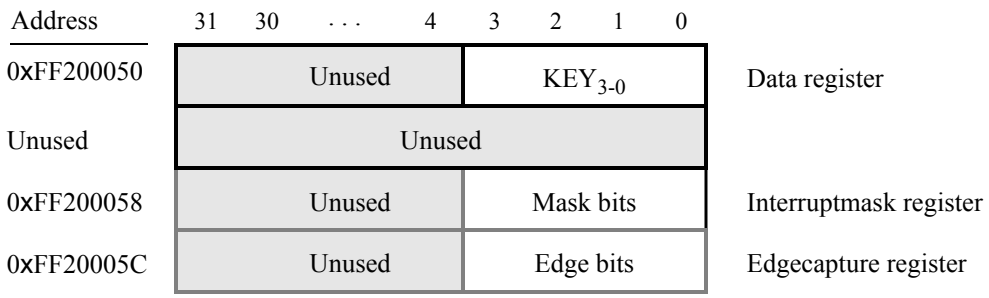


Figure 6. Registers used in the pushbutton parallel port.

2.3.5 Expansion Parallel Port

The *DE1-SoC Computer with Nios II* includes two bidirectional parallel ports that are connected to the *JP1* and *JP2* 40-pin headers on the DE1-SoC board. These parallel ports include the four 32-bit registers that were described previously for Figure 2. The base address of the port for JP1 is 0xFF200060, and for JP2 is 0xFF200070. Figure 7 gives a diagram of the 40-pin connectors on the DE1-SoC board, and shows how the respective parallel port *Data* register bits, D_{31-0}, are assigned to the pins on the connector. The figure shows that bit D_0 of the parallel port is assigned to the pin at the top right corner of the connector, bit D_1 is assigned below this, and so on. Note that some of the pins on the 40-pin header are not usable as input/output connections, and are therefore not used by the parallel ports. Also, only 32 of the 36 data pins that appear on each connector can be used.

2.3.6 Using the Parallel Ports with Assembly Language Code and C Code

The *DE1-SoC Computer with Nios II* provides a convenient platform for experimenting with Nios II assembly language code, or C code. A simple example of such code is provided in the Appendix in Listings 1 and 2. Both

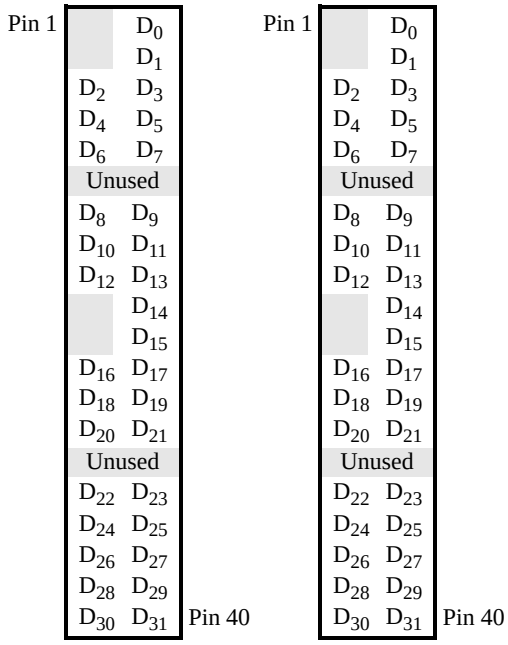


Figure 7. Assignment of parallel port bits to pins on *JP1* and *JP2*.

programs perform the same operations, and illustrate the use of parallel ports by using either assembly language or C code. Each listing *includes* a file that specifies the memory-mapped addresses of all peripheral devices in the *DE1-SoC Computer with Nios II*. These include files, called *address_map_nios2.s* and *address_map_nios2.h*, are provided in Listings 3 and 4. These include files are also used in other code samples described in this document.

The code in Listing 1 and 2 displays the values of the SW switches on the LED lights, and also shows a rotating pattern on the LEDs. This pattern is shifted in a loop, using a software delay to make the shifting slow enough to observe. The pattern can be changed to the values of the SW switches by pressing a pushbutton KEY. When a KEY is pressed, the program waits in a loop until it is released and then continues to display the pattern.

2.4 JTAG* Port

The JTAG* port implements a communication link between the DE1-SoC board and its host computer. This link can be used by the Altera Quartus® Prime software to transfer FPGA programming files into the DE1-SoC board, and by software development tools that are used to develop and debug Nios II programs. The JTAG port also includes a UART, which can be used to transfer character data between the host computer and programs that are executing on the Nios II processor. The programming interface of the JTAG UART consists of two 32-bit registers, as shown in Figure 8. The register mapped to address `0xFF201000` is called the *Data* register and the register mapped to address `0xFF201004` is called the *Control* register.

When character data from the host computer is received by the JTAG UART it is stored in a 64-character FIFO. The number of characters currently stored in this FIFO is indicated in the field *RAVAIL*, which are bits 31−16 of the *Data* register. If the receive FIFO overflows, then additional data is lost. When data is present in the receive FIFO,

Address	31 · · · 16	15	14 · · · 11	10	9	8	7 · · · 1	0		
0xFF201000	RAVAIL	RVALID	Unused				DATA			Data register
0xFF201004	WSPACE	Unused		AC	WI	RI		WE	RE	Control register

Figure 8. JTAG UART registers.

then the value of *RAVAIL* will be greater than 0 and the value of bit 15, *RVALID*, will be 1. Reading the character at the head of the FIFO, which is provided in bits $7-0$, decrements the value of *RAVAIL* by one and returns this decremented value as part of the read operation. If no data is present in the receive FIFO, then *RVALID* will be set to 0 and the data in bits $7-0$ is undefined.

The JTAG UART also includes a 64-character FIFO that stores data waiting to be transmitted to the host computer. Character data is loaded into this FIFO by performing a write to bits $7-0$ of the *Data* register in Figure 8. Note that writing into this register has no effect on received data. The amount of space, *WSPACE*, currently available in the transmit FIFO is provided in bits $31-16$ of the *Control* register. If the transmit FIFO is full, then any characters written to the *Data* register will be lost.

Bit 10 in the *Control* register, called *AC*, has the value 1 if the JTAG UART has been accessed by the host computer. This bit can be used to check if a working connection to the host computer has been established. The *AC* bit can be cleared to 0 by writing a 1 into it.

The *Control* register bits *RE*, *WE*, *RI*, and *WI* are described in Section 3.

2.4.1 Using the JTAG* UART with Assembly Language Code and C Code

Listings 5 and 6 give simple examples of assembly language and C code, respectively, that use the JTAG UART. Both versions of the code perform the same function, which is to first send an ASCII string to the JTAG UART, and then enter an endless loop. In the loop, the code reads character data that has been received by the JTAG UART, and echoes this data back to the UART for transmission. In the *CPUlator* simulator, there is a JTAG window that allows text to be typed and echoed. If the program is executed in a hardware system, then the software developmemt tools being used would provide a *Terminal Window* that is used to display any keyboard characters typed.

2.5 Interval Timers

The *DE1-SoC Computer with Nios II* includes a timer module implemented in the FPGA that can be used by the Nios II processor. This timer can be loaded with a preset value, and then counts down to zero using a 100-MHz clock. The programming interface for the timer includes six 16-bit registers, as illustrated in Figure 9. The 16-bit register at address 0xFF202000 provides status information about the timer, and the register at address 0xFF202004 allows control settings to be made. The bit fields in these registers are described below:

- *TO* provides a timeout signal which is set to 1 by the timer when it has reached a count value of zero. The *TO* bit can be reset by writing a 0 into it.

- *RUN* is set to 1 by the timer whenever it is currently counting. Write operations to the status halfword do not affect the value of the *RUN* bit.
- *ITO* is used for generating interrupts, which are discussed in section 3.

Address	31 . . . 17 16	15 . . . 3	2	1	0	
0xFF202000	Not present (interval timer has 16-bit registers)	Unused		RUN	TO	Status register
0xFF202004		Unused	STOP START	CONT	ITO	Control register
0xFF202008		Counter start value (low)				
0xFF20200C		Counter start value (high)				
0xFF202010		Counter snapshot (low)				
0xFF202014		Counter snapshot (high)				

Figure 9. Interval timer registers.

- *CONT* affects the continuous operation of the timer. When the timer reaches a count value of zero it automatically reloads the specified starting count value. If *CONT* is set to 1, then the timer will continue counting down automatically. But if *CONT* = 0, then the timer will stop after it has reached a count value of 0.
- (*START*/*STOP*) is used to commence/suspend the operation of the timer by writing a 1 into the respective bit.

The two 16-bit registers at addresses 0xFF202008 and 0xFF20200C allow the period of the timer to be changed by setting the starting count value. The default setting gives a timer period of 125 msec. To achieve this period, the starting value of the count is 100 MHz × 125 msec = 12.5×10^6. It is possible to capture a snapshot of the counter value at any time by performing a write to address 0xFF202010. This write operation causes the current 32-bit counter value to be stored into the two 16-bit timer registers at addresses 0xFF202010 and 0xFF202014. These registers can then be read to obtain the count value.

A second interval timer, which has an identical interface to the one described above, is also available in the FPGA, starting at the base address 0xFF202020.

Each Nios II processor has exclusive access to two interval timers.

2.6 G-Sensor

The *DE1-SoC Computer with Nios II* includes a 3D accelerometer (G-sensor) that is connected to the HPS. The Nios II processor can access this device via an I2C serial interface at the base address 0xFFC04000. More details can be found in the tutorial *Using the Accelerometer on DE-series Boards*.

2.7 Floating-point Hardware

The Nios II processor in the DE1-SoC Computer includes hardware support for floating-point addition, subtraction, multiplication, and division. To use this support in a C program, variables must be declared with the type *float*. A simple example of such code is given in Listing 19. When this code is compiled, it is necessary to pass the special argument -mcustom-fpu-cfg=60-2 to the C compiler, to instruct it to use the floating-point hardware support.

2.8 System ID

The system ID module provides a unique value that identifies the DE1-SoC Computer system. The host computer connected to the DE1-SoC board can query the system ID module by performing a read operation through the JTAG port. The host computer can then check the value of the returned identifier to confirm that the DE1-SoC Computer has been properly downloaded onto the DE1-SoC board. This process allows debugging tools on the host computer, such as the Monitor Program, to verify that the DE1-SoC board contains the required computer system before attempting to execute code that has been compiled for this system.

3 Exceptions and Interrupts

The reset address of the Nios II processor in the DE1-SoC Computer is set to 0x00000000. The address used for all other general exceptions, such as divide by zero, and hardware IRQ interrupts is 0x00000020. Since the Nios II processor uses the same address for general exceptions and hardware IRQ interrupts, the Exception Handler software must determine the source of the exception by examining the appropriate processor status register. Table 1 gives the assignment of IRQ numbers to each of the I/O peripherals in the DE1-SoC Computer. The rest of this section describes the interrupt behavior associated with the interval timer, parallel ports, and serial ports in the DE1-SoC Computer.

I/O Peripheral	IRQ #
Interval timer	0
Pushbutton switch parallel port	1
Second Interval timer	2
Audio port	6
PS/2 port	7
JTAG port	8
IrDA port	9
Serial port	10
JP1 Expansion parallel port	11
JP2 Expansion parallel port	12
PS/2 port dual	23

Table 1. Hardware IRQ interrupt assignment for the DE1-SoC Computer.

3.1 Interrupts from Parallel Ports

Parallel ports were illustrated in Figure 2, which is reproduced as Figure 10. As the figure shows, parallel ports that support interrupts include two related registers at the addresses *Base* + 8 and *Base* + C. The *Interruptmask* register, which has the address *Base* + 8, specifies whether or not an interrupt signal should be sent to the processor when the data present at an input port changes value. Setting a bit location in this register to 1 allows interrupts to be generated, while setting the bit to 0 prevents interrupts. Finally, the parallel port may contain an *Edgecapture* register at address *Base* + C. Each bit in this register has the value 1 if the corresponding bit location in the parallel port has changed its value from 0 to 1. A bit in the *Edgecapture* register can be cleared to 0 by writing a 1 into the corresponding bit position, which clears any associated interrupt.

Address	31 30 · · · 4 3 2 1 0	
Base	Input or output data bits	Data register
Base + 4	Direction bits	Direction register
Base + 8	Mask bits	Interruptmask register
Base + C	Edge bits	Edgecapture register

Figure 10. Registers used for interrupts from the parallel ports.

3.1.1 Interrupts from the Pushbutton KEY Port

Figure 6, reproduced as Figure 11, shows the registers associated with the pushbutton KEY port. The *Interruptmask* register allows interrupts to be generated when a key is pressed. Interrupts can be enabled individually for each key by setting its *Interruptmask* bit to 1. When a key is pressed, the corresponding bit in the *Edgecapture* register is set to 1 by the parallel port. This bit remains 1 until cleared to 0 by software. An interrupt service routine can read the *Edgecapture* register to determine which key/s has/have been pressed. An *Edgecapture* register bit can be cleared by writing a logic value 1 into the bit position. Clearing the bit resets the corresponding interrupt signal being sent to the processor.

Address	31 30 · · · 4	3 2 1 0	
0xFF200050	Unused	KEY_{3-0}	Data register
Unused	Unused		
0xFF200058	Unused	Mask bits	Interruptmask register
0xFF20005C	Unused	Edge bits	Edgecapture register

Figure 11. Registers used for interrupts from the pushbutton KEY port.

3.2 Interrupts from the JTAG* UART

Figure 8, reproduced as Figure 12, shows the *Data* and *Control* registers of the JTAG UART. As we said in Section 2.4, *RAVAIL* in the *Data* register gives the number of characters that are stored in the receive FIFO, and *WSPACE* gives the amount of unused space that is available in the transmit FIFO. The *RE* and *WE* bits in Figure 12 are used to enable processor interrupts associated with the receive and transmit FIFOs. When enabled, interrupts are generated when *RAVAIL* for the receive FIFO, or *WSPACE* for the transmit FIFO, exceeds 7. Pending interrupts are indicated in the Control register's *RI* and *WI* bits, and can be cleared by writing or reading data to/from the JTAG UART.

Address	31 · · · 16	15	14 · · · 11	10	9	8	7 · · · 1	0	
0xFF201000	RAVAIL	RVALID	Unused				DATA		Data register
0xFF201004	WSPACE	Unused		AC	WI	RI		WE	RE	Control register

Figure 12. Interrupt bits in the JTAG UART registers.

3.3 Interrupts from the FPGA Interval Timer

Figure 9, in Section 2.5, shows six registers that are associated with the interval timer. As we said in Section 2.5, the *TO* bit in the *Status* register is set to 1 when the timer reaches a count value of 0. It is possible to generate an interrupt when this occurs, by using the *ITO* bit in the *Control* register. Setting the *ITO* bit to 1 causes an interrupt request to be sent to the processor whenever *TO* becomes 1. After an interrupt occurs, it can be cleared by writing any value into the *Status* register.

3.4 Using Interrupts with Assembly Language Code

An example of assembly language code for the DE1-SoC Computer that uses interrupts is shown in Listing 7. When this code is executed on the DE1-SoC board it displays a rotating pattern on the LEDs. The pattern's rotation can be toggled through pressing the pushbutton KEYs. Two types of interrupts are used in the code. The LEDs are controlled by an interrupt service routine for the interval timer, and another interrupt service routine is used to handle the pushbutton keys. The speed of the rotation is set in the main program, by using a counter value in the interval timer that causes an interrupt to occur every 50 msec.

The reset and exception handlers for the main program in Listing 7 are given in Listing 8. The reset handler simply jumps to the *_start* symbol in the main program. The exception handler first checks if the exception that has occurred is an external interrupt or an internal one. In the case of an internal exception, such as an illegal instruction opcode or a trap instruction, the handler simply exits, because it does not handle these cases. For external exceptions, it calls either the interval timer interrupt service routine, for a level 0 interrupt, or the pushbutton key interrupt service routine for level 1. These routines are shown in Listings 9 and 10, respectively.

3.5 Using Interrupts with C Language Code

An example of C language code for the DE1-SoC Computer that uses interrupts is shown in Listing 11. This code performs exactly the same operations as the code described in Listing 7.

To enable interrupts the code in Listing 11 uses *macros* that provide access to the Nios II status and control registers. A collection of such macros, which can be used in any C program, are provided in Listing 12.

The reset and exception handlers for the main program in Listing 11 are given in Listing 13. The function called *the_reset* provides a simple reset mechanism by performing a branch to the main program. The function named *the_exception* represents a general exception handler that can be used with any C program. It includes assembly language code to check if the exception is caused by an external interrupt, and, if so, calls a C language routine named *interrupt_handler*. This routine can then perform whatever action is needed for the specific application. In Listing 13, the *interrupt_handler* code first determines which exception has occurred, by using a macro from Listing 12 that reads the content of the Nios II interrupt pending register. The interrupt service routine that is invoked for the interval timer is shown in 14, and the interrupt service routine for the pushbutton switches appears in Listing 15.

The source code files shown in Listing 7 to Listing 15 are distributed as part of the Monitor Program. The files can be found under the heading *sample programs*, and are identified by the name *Interrupt Example*.

4 Media Components

This section describes the audio in/out, video-out, video-in, PS/2, IrDA*, and ADC ports.

4.1 Audio In/Out Port

The *DE1-SoC Computer with Nios II* includes an audio port that is connected to the audio CODEC (COder/DECoder) chip on the DE1-SoC board. The default setting for the sample rate provided by the audio CODEC is 8K samples/sec. The audio port provides audio-input capability via the microphone jack on the DE1-SoC board, as well as audio output functionality via the line-out jack. The audio port includes four FIFOs that are used to hold incoming and outgoing data. Incoming data is stored in the left- and right-channel *Read* FIFOs, and outgoing data is held in the left- and right-channel *Write* FIFOs. All FIFOs have a maximum depth of 128 32-bit words.

The audio port's programming interface consists of four 32-bit registers, as illustrated in Figure 13. The *Control* register, which has the address `0xFF203040`, is readable to provide status information and writable to make control settings. Bit *RE* of this register provides an interrupt enable capability for incoming data. Setting this bit to 1 allows the audio core to generate a Nios II interrupt when either of the *Read* FIFOs are filled 75% or more. The bit *RI* will then be set to 1 to indicate that the interrupt is pending. The interrupt can be cleared by removing data from the *Read* FIFOs until both are less than 75% full. Bit *WE* gives an interrupt enable capability for outgoing data. Setting this bit to 1 allows the audio core to generate an interrupt when either of the *Write* FIFOs are less that 25% full. The bit *WI* will be set to 1 to indicate that the interrupt is pending, and it can be cleared by filling the *Write* FIFOs until both are more than 25% full. The bits *CR* and *CW* in Figure 13 can be set to 1 to clear the *Read* and *Write* FIFOs, respectively. The clear function remains active until the corresponding bit is set back to 0.

The read-only *Fifospace* register in Figure 13 contains four 8-bit fields. The fields *RARC* and *RALC* give the number of words currently stored in the right and left audio-input FIFOs, respectively. The fields *WSRC* and *WSLC* give the number of words currently available (that is, *unused*) for storing data in the right and left audio-out FIFOs. When all

Address	31 . . . 24	23 . . . 16	15 . . . 10	9	8	7 . . . 4	3	2	1	0	
0xFF203040	Unused			WI	RI		CW	CR	WE	RE	Control
0xFF203044	WSLC	WSRC	RALC			RARC					Fifospace
0xFF203048	Left data										Leftdata
0xFF20304C	Right data										Rightdata

Figure 13. Audio port registers.

FIFOs in the audio port are cleared, the values provided in the *Fifospace* register are *RARC* = *RALC* = 0 and *WSRC* = *WSLC* = 128.

The *Leftdata* and *Rightdata* registers are readable for audio in, and writable for audio out. When data is read from these registers, it is provided from the head of the *Read* FIFOs, and when data is written into these registers it is loaded into the *Write* FIFOs.

A fragment of C code that uses the audio port is shown in Listing 16. The code checks to see when the depth of either the left or right *Read* FIFO has exceeded 75% full, and then moves the data from these FIFOs into a memory buffer.

4.2 Video-out Port

The *DE1-SoC Computer with Nios II* includes a video-out port connected to the on-board VGA controller that can be connected to a standard VGA monitor. The video-out port support a screen resolution of 640 × 480. The image that is displayed by the video-out port is derived from two sources: a *pixel* buffer, and a *character* buffer.

4.2.1 Pixel Buffer

The pixel buffer for the video-out port holds the data (color) for each pixel that will be displayed. As illustrated in Figure 14, the pixel buffer provides an image resolution of 320 × 240 pixels, with the coordinate 0,0 being at the top-left corner of the image. Since the video-out port supports the screen resolution of 640 × 480, each of the pixel values in the pixel buffer is replicated in both the *x* and *y* dimensions when it is being displayed on the screen.

Figure 15*a* shows that each pixel color is represented as a 16-bit halfword, with five bits for the blue and red components, and six bits for green. As depicted in part *b* of Figure 15, pixels are addressed in the pixel buffer by using the combination of a *base* address and an *x,y* offset. In the DE1-SoC Computer the default address of the pixel buffer is 0x08000000, which corresponds to the starting address of the FPGA on-chip memory. Using this scheme, the pixel at location 0,0 has the address 0x08000000, the pixel 1,0 has the address *base* + $(00000000\ 000000001\ 0)_2$ = 0x08000002, the pixel 0,1 has the address *base* + $(00000001\ 000000000\ 0)_2$ = 0x08000400, and the pixel at location 319,239 has the address *base* + $(11101111\ 100111111\ 0)_2$ = 0x0803BE7E.

You can create an image by writing color values into the pixel addresses as described above. A dedicated *pixel buffer controller* continuously reads this pixel data from sequential addresses in the corresponding memory for display on

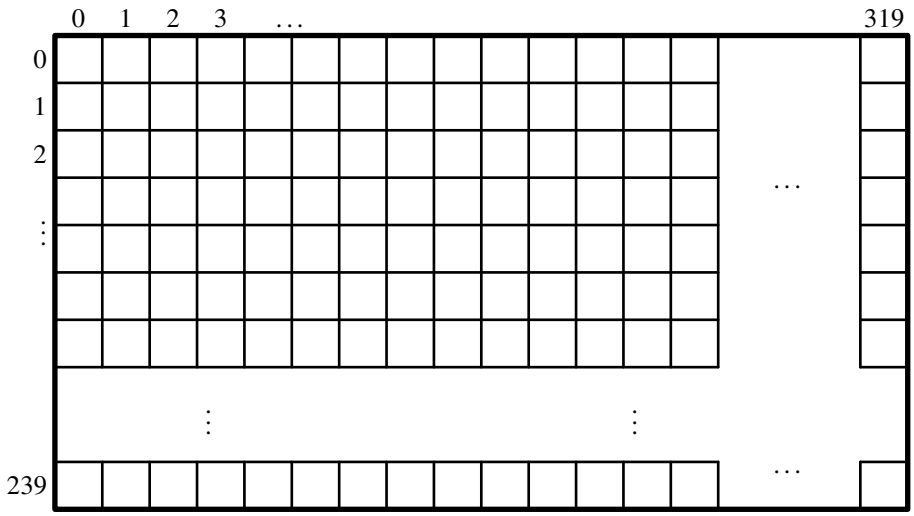
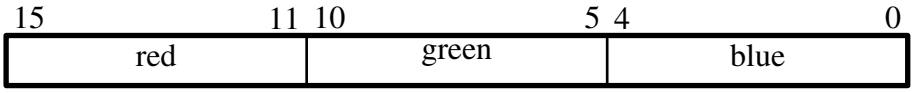
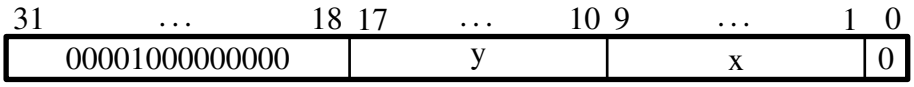


Figure 14. Pixel buffer coordinates.



(a) Pixel values



(b) Pixel address

Figure 15. Pixel values and addresses.

the screen. You can modify the pixel data at any time, simply by writing to the pixel addresses. Thus, an image can be changed even when it is in the process of being displayed. However, it is also possible to avoid making changes to the pixel buffer while it is being displayed, by using the concept of *double-buffering*. In this scheme, two pixel buffers are involved, called the *front* and *back* buffers, described below.

4.2.2 RGB Resampling

The DE1-SoC Computer contains an RGB Resampler for converting video streams between RGB color formats. Reading from the 32-bit *Status* register at address `0xFF203010` provides information about alpha/no alpha, color/grayscale, and mode for the incoming and outgoing formats. The incoming format for the DE1-SoC Computer video stream is `0x14`, which corresponds to no alpha, color, 16-bit RGB (5-bit Red, 6-bit Green, 5-bit Blue). More information can be found in the documentation called *Video IP Cores for Altera DE-Series Computer Systems*, available as part of the `Hardware Components` tutorials on the FPGAcademy.org website.

4.2.3 Double Buffering

As mentioned above, a pixel buffer controller reads data out of the pixel buffer so that it can be displayed on the screen. This pixel buffer controller includes a programming interface in the form of a set of registers, as illustrated in Table 2. The register at address 0xFF203020 is called the *Buffer* register, and the register at address 0xFF203024 is the *Backbuffer* register. Each of these registers stores the starting address of a pixel buffer. The Buffer register holds the address of the pixel buffer that is displayed on the screen. As mentioned above, in the default configuration of the DE1-SoC Computer this Buffer register is set to the address 0x08000000, which points to the start of the FPGA on-chip memory. The default value of the Backbuffer register is also 0x08000000, which means that there is only one pixel buffer. But software can modify the address stored in the Backbuffer register, thereby creating a second pixel buffer. The pixel buffer can be located in the SDRAM memory in the DE1-SoC Computer, which has the base address 0x00000000. Note that the pixel buffer cannot be located in the DDR3 memory in the DE1-SoC Computer, because the pixel buffer controller is not connected to the DDR3 memory. An image can be drawn into the second buffer by writing to its pixel addresses. This image is not displayed on the screen until a pixel buffer *swap* is performed, as explained below.

A pixel buffer swap is caused by writing the value 1 to the Buffer register. This write operation does not directly modify the content of the Buffer register, but instead causes the contents of the Buffer and Backbuffer registers to be swapped. The swap operation does not happen right away; it occurs at the end of a screen-drawing cycle, after the last pixel in the bottom-right corner has been displayed. This time instance is referred to as the *vertical synchronization* time, and occurs every 1/60 seconds. Software can poll the value of the *S* bit in the *Status* register, at address 0xFF20302C, to see when the vertical synchronization has happened. Writing the value 1 into the Buffer register causes *S* to be set to 1. Then, when the swap of the Buffer and Backbuffer registers has been completed *S* is reset back to 0.

Address	Register Name	R/W	Bit Description								
			31…24	23…16	15…12	11…8	7…6	5…3	2	1	0
0xFF203020	`Buffer`	R	Buffer's start address								
0xFF203024	`BackBuffer`	R/W	Back buffer's start address								
0xFF203028	`Resolution`	R	Y		X						
0xFF20302C	`Status`	R	m	n	*(1)*	BS	SB	*(1)*	EN	A	S
	`Control`	W	*(1)*						EN	*(1)*	

Notes:

(1) Reserved. Read values are undefined. Write zero.

Table 2. Pixel Buffer Controller

In a typical application the pixel buffer controller is used as follows. While the image contained in the pixel buffer that is pointed to by the Buffer register is being displayed, a new image is drawn into the pixel buffer pointed to by the Backbuffer register. When this new image is ready to be displayed, a pixel buffer swap is performed. Then, the pixel buffer that is now pointed to by the Backbuffer register, which was already displayed, is cleared and the next new image is drawn. In this way, the next image to be displayed is always drawn in the "back" pixel buffer, and the two pixel buffer pointers are swapped when the new image is ready to be displayed. Each time a swap is performed software has to synchronize with the video-out port by waiting until the *S* bit in the Status register becomes 0.

As shown in Table 2 the *Status* register contains additional information other than the S bit. The fields n and m give the number of address bits used for the X and Y pixel coordinates, respectively. The BS field specifies the number of data bits per symbol minus one. The SB field specifies the number of symbols per beat minus one. The A field allows the selection of two different ways of forming pixel addresses. If configured with $A = 0$, then the pixel controller expects addresses to contain X and Y fields, as we have used in this section. But if $A = 1$, then the controller expects addresses to be consecutive values starting from 0 and ending at the total number of pixels−1. The EN field is used to enable or disable the DMA controller. If this bit is set to 0, the DMA controller will be turned off.

In Table 2 the default values of the status register fields in the DE1-SoC Computer are used when forming pixel addresses. The defaults are $n = 9$, $m = 8$, and $A = 0$. If the pixel buffer controller is changed to provide different values of these fields, then the way in which pixel addresses are formed has to be modified accordingly. The programming interface also includes a *Resolution* register, shown in Table 2, that contains the X and Y resolution of the pixel buffer(s).

4.2.4 Character Buffer

The character buffer for the video-out port is stored in on-chip memory in the FPGA on the DE1-SoC board. As illustrated in Figure 16*a*, the buffer provides a resolution of 80 × 60 characters, where each character occupies an 8 × 8 block of pixels on the screen. Characters are stored in each of the locations shown in Figure 16*a* using their ASCII codes; when these character codes are displayed on the monitor, the character buffer automatically generates the corresponding pattern of pixels for each character using a built-in font. Part *b* of Figure 16 shows that characters are addressed in the memory by using the combination of a *base* address, which has the value 0x09000000, and an *x,y* offset. Using this scheme, the character at location 0,0 has the address 0x09000000, the character 1,0 has the address *base* + $(000000\ 0000001)_2$ = 0x09000001, the character 0,1 has the address *base* + $(000001\ 0000000)_2$ = 0x09000080, and the character at location 79,59 has the address *base* + $(111011\ 1001111)_2$ = 0x09001DCF.

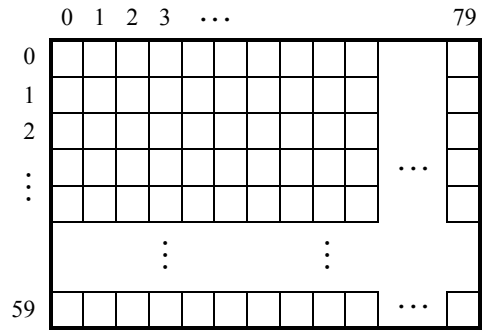
4.2.5 Using the Video-out Port with C code

A fragment of C code that uses the pixel and character buffers is shown in Listing 17. The first **for** loop in the figure draws a rectangle in the pixel buffer using the color *pixel_color*. The rectangle is drawn using the coordinates x_1, y_1 and x_2, y_2. The second **while** loop in the figure writes a null-terminated character string pointed to by the variable *text_ptr* into the character buffer at the coordinates x, y.

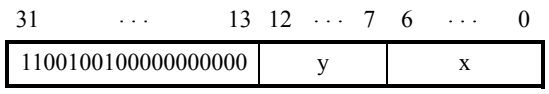
4.3 Video-in Port

The *DE1-SoC Computer with Nios II* includes a video-in port for use with the composite video-in connector on the DE1-SoC board. The video analog-to-digital converter (ADC) connected to this port is configured to support an NTSC video source. The video-in port provides frames of video at a resolution of 320 x 240 pixels. These video frames can be displayed on a monitor by using the video-out port described in Section 4.2. The video-in port writes each frame of the video-in data into the pixel buffer described in Section 4.2.1. The video-in port can be configured to provide two types of images: either the "raw" image provided by the video ADC, or a version of this image in which only "edges" that are detected in the image are drawn.

The video-in port has a programming interface that consists of two registers, as illustrated in Figure 17. The *Control* register at the address 0xFF20306C is used to enable or disable the video input. If the *EN* bit in this register is set



(a) Character buffer coordinates



(b) Character buffer addresses

Figure 16. Character buffer coordinates and addresses.

to 0, then the video-in core does not store any data into the pixel buffer. Setting *EN* to 1 and then changing *EN* to 0 can be used to capture a still picture from the video-in port.

The register at address 0xFF203070 is used to enable or disable edge detection. Setting the *E* bit in this register to 1 causes the input video to passed through hardware circuits that detect edges in the images. The image stored in the pixel buffer will then consist of dark areas that are punctuated by lighter lines along the edges that have been detected. Setting $E = 0$ causes a normal image to be stored into the pixel buffer.

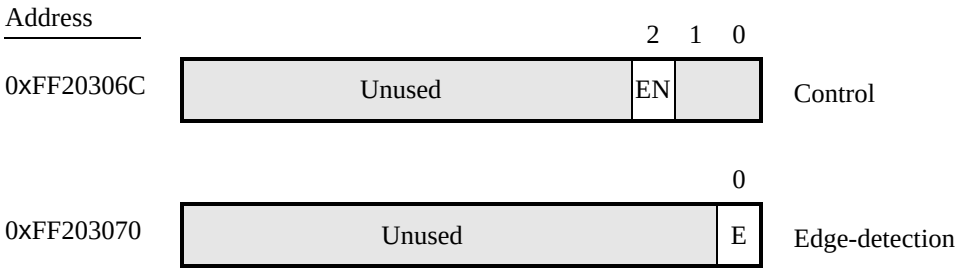


Figure 17. The video-in port programming interface.

4.3.1 DMA Controller for Video

The data provided by the Video-In core is stored into memory using a DMA Controller for Video. When operating in *Stream to Memory* mode, the DMA stores the incoming frames to memory. Table 3 describes the registers used in the DMA Controller.

Address	Register Name	R/W	Bit Description								
			31…24	23…16	15…12	11…8	7…6	5…3	2	1	0
0xFF203060	`Buffer`	R	Buffer's start address								
0xFF203064	`BackBuffer`	R/W	Back buffer's start address								
0xFF203068	`Resolution`	R	Y		X						
0xFF20306C	`Status`	R	m	n	*(1)*	BS	SB	*(1)*	EN	A	S
	`Control`	W	*(1)*						EN	*(1)*	

Notes:
(1) Reserved. Read values are undefined. Write zero.

Table 3. Video DMA Controller

The incoming video is stored to memory, starting at the address specified in the *Buffer* register. The *BackBuffer* register is used to store an alternate memory location. To change where the video is stored, the new location should first be written into the *BackBuffer*. Then the value in the *BackBuffer* and *Buffer* registers can be switched by performing a write to the *Buffer* register.

Bit 2 of the *Status/Control* register, *EN*, is used to enable or disable the Video DMA controller. In the DE1-SoC Computer, the DMA controller is disabled by default. To enable the DMA controller, write a 1 into this location. The Video DMA Controller will then begin storing the video into the location specified in the *Buffer* register.

The default value stored in the *Buffer* register is `0x08000000`. This address is also used as the source for the Video-Out port, as described in Section 4.2, allowing the Video In stream to be displayed on the VGA. If the Video-Out is intended to display a different signal, than the address stored in the Video DMA Controller's *Buffer* register should be changed.

4.4 Audio/Video Configuration Module

The audio/video configuration module controls settings that affect the operation of both the audio port and the video-out port. The audio/video configuration module automatically configures and initializes both of these ports whenever the *DE1-SoC Computer with Nios II* is reset. For typical use of the *DE1-SoC Computer with Nios II* it is not necessary to modify any of these default settings.

4.5 PS/2 Port

The *DE1-SoC Computer with Nios II* includes two PS/2 ports that can be connected to a standard PS/2 keyboard or mouse. The port includes a 256-byte FIFO that stores data received from a PS/2 device. The programming interface for the PS/2 port consists of two registers, as illustrated in Figure 18. The *PS2_Data* register is both readable and writable. When bit 15, *RVALID*, is 1, reading from this register provides the data at the head of the FIFO in the *Data* field, and the number of entries in the FIFO (including this read) in the *RAVAIL* field. When *RVALID* is 1, reading from the *PS2_Data* register decrements this field by 1. Writing to the *PS2_Data* register can be used to send a command in the *Data* field to the PS/2 device.

The *PS2_Control* register can be used to enable interrupts from the PS/2 port by setting the *RE* field to the value 1. When this field is set, then the PS/2 port generates an interrupt when *RAVAIL* > 0. While the interrupt is pending the field *RI* will be set to 1, and it can be cleared by emptying the PS/2 port FIFO. The *CE* field in the *PS2_Control* register is used to indicate that an error occurred when sending a command to a PS/2 device.

Address	31 · · · 16	15	· · · 10	9	8	7 · · · 1	0	
0xFF200100	RAVAIL	RVALID	Unused			Data		PS2_Data
0xFF200104			CE		RI		RE	PS2_Control

Figure 18. PS/2 port registers.

A fragment of C code that uses the PS/2 port is given in Listing 18. This code reads the content of the *Data* register, and saves data when it is available. If the code is used continually in a loop, then it stores the last three bytes of data received from the PS/2 port in the variables $byte_1$, $byte_2$, and $byte_3$.

4.5.1 PS/2 Port Dual

A second PS/2 port is included that allows both a keyboard and mouse to be used at the same time. To use the dual port a Y-splitter cable must be used and the keyboard and mouse must be connected to the PS/2 connector on the DE1-SoC board through this cable. The PS/2 port dual has the same registers as the PS/2 port shown in Listing 18, except that the base address of its *PS2_Data* register is 0xFF200108 and the base address of its *PS2_Control* register is 0xFF20010C.

4.6 IrDA* Infrared Serial Port

The IrDA port in the *DE1-SoC Computer with Nios II* implements a UART that is connected to the infrared transmit/receive device on the DE1-SoC board. This UART is configured for 8-bit data, one stop bit, and no parity, and operates at a baud rate of 115,200. The serial port's programming interface consists of two 32-bit registers, as illustrated in Figure 19. The register at address 0xFF201020 is referred to as the *Data* register, and the register at address 0xFF201024 is called the *Control* register.

Address	31 · · · 24	23 · · · 16	15	14 · · · 10	9	8	7 · · · 1		0	
0xFF201020	Unused	RAVAIL	RVALID	Unused	PE		DATA			Data register
0xFF201024	Unused	WSPACE	Unused		WI	RI		WE	RE	Control register

Figure 19. IrDA serial port UART registers.

When character data is received from the IrDA chip it is stored in a 256-character FIFO in the UART. As illustrated in Figure 19, the number of characters *RAVAIL* currently stored in this FIFO is provided in bits 23−16 of the *Data* register. If the receive FIFO overflows, then additional data is lost. When the data that is present in the receive FIFO is available for reading, then the value of bit 15, *RVALID*, will be 1. Reading the character at the head of the FIFO,

which is provided in bits 7 – 0, decrements the value of *RAVAIL* by one and returns this decremented value as part of the read operation. If no data is available to be read from the receive FIFO, then *RVALID* will be set to 0 and the data in bits 7 – 0 is undefined.

The UART also includes a 256-character FIFO that stores data waiting to be sent to the IrDA device. Character data is loaded into this register by performing a write to bits 7–0 of the *Data* register. Writing into this register has no effect on received data. The amount of space *WSPACE* currently available in the transmit FIFO is provided in bits 23–16 of the *Control* register, as indicated in Figure 19. If the transmit FIFO is full, then any additional characters written to the *Data* register will be lost.

The *RE* and *WE* bits in the *Control* register are used to enable Nios II processor interrupts associated with the receive and transmit FIFOs. When enabled, interrupts are generated when *RAVAIL* for the receive FIFO, or *WSPACE* for the transmit FIFO, exceeds 31. Pending interrupts are indicated in the *Control* register's *RI* and *WI* bits, and can be cleared by writing or reading data to/from the UART.

4.7 Analog-to-Digital Conversion Port

The Analog-to-Digital Conversion (ADC) Port provides access to the eight-channel, 12-bit analog-to-digital converter on the DE1-SoC board. As illustrated in Figure 20, the ADC port comprises eight 12-bit registers starting at the base address 0xFF204000. The first two registers have dual purposes, acting as both data and control registers. By default, the ADC port updates the A-to-D conversion results for all ports only when instructed to do so. Writing to the control register at address 0xFF204000 causes this update to occur. Reading from the register at address 0xFF204000 provides the conversion data for channel 0. Reading from the other seven registers provides the conversion data for the corresponding channels. It is also possible to have the ADC port continually request A-to-D conversion data for all channels. This is done by writing the value 1 to the control register at address 0xFF204004. The *R* bit of each channel register in Figure 20 is used in Auto-update mode. *R* is set to 1 when its corresponding channel is refreshed and set to 0 when the channel is read.

Address	31 . . . 16	15	14 12	11 . . . 0	
0xFF204000	Unused	R	Unused		Channel 0 / Update
0xFF204004	Unused	R	Unused		Channel 1 / Auto-update
0xFF204008	Unused	R	Unused		Channel 2
	... not shown				
0xFF20401C	Unused	R	Unused		Channel 7

Figure 20. ADC port registers.

Figure 21 shows the connector on the DE1-SoC board that is used with the ADC port. Analog signals in the range of 0 V to the V_{CC5} power-supply voltage can be connected to the pins for channels 0 to 7.

Gnd	Ch_7
Ch_6	Ch_5
Ch_4	Ch_3
Ch_2	Ch_1
Ch_0	V_{cc5}

JP15

Figure 21. ADC connector.

5 Modifying the DE1-SoC Computer with Nios II

It is possible to modify the *DE1-SoC Computer with Nios II* by using Altera's Quartus® Prime software and Platform Designer tool. Instructions for using this software are provided as part of the `Computer Organization and System Design` tutorials on the FPGAcademy.org website. To modify the system it is first necessary to make an editable copy of the *DE1-SoC Computer with Nios II*. The files for this system are available in its GitHub repository.

Locate these files, copy them to a working directory, and then use the Quartus Prime and Platform Designer software to make any desired changes.

Table 4 lists the names of the Platform Designer IP cores that are used in this system. When the *DE1-SoC Computer with Nios II* design files are opened in the Quartus Prime software, these cores can be examined using the Platform Designer System Integration tool. Each core has a number of settings that are selectable in the Platform Designer System Integration tool, and includes a datasheet that provides detailed documentation.

The steps needed to modify the system are:

1. Make of copy of the design source files for the *DE1-SoC Computer with Nios II* from the its GitHub repository.
2. Open the top-level project file (*.*qpf*) in the Quartus Prime software
3. Open the Platform Designer System Integration tool in the Quartus Prime software, and modify the system as desired
4. Generate the modified system by using the Platform Designer System Integration tool
5. It may be necessary to modify the Verilog code in the top-level module of the project, if any I/O peripherals have been added or removed from the system
6. Compile the project in the Quartus Prime software
7. Download the modified system into the DE1-SoC board

Note: to compile and use a new version of the *DE1-SoC Computer with Nios II* it may be necessary to request a license from Altera that allows you to create circuit that includes the Nios II processor.

I/O Peripheral	Qsys Core
SDRAM	SDRAM Controller
SRAM	SRAM Controller
On-chip memory character buffer	Character Buffer for VGA Display
SD Card	SD Card Interface
Flash	Flash Memory IP Core
Red LED parallel port	Parallel Port
7-segment displays parallel port	Parallel Port
Expansion parallel ports	Parallel Port
Slider switch parallel port	Parallel Port
Pushbutton parallel port	Parallel Port
PS/2 port	PS2 Controller
JTAG port	JTAG UART
Serial port	RS232 UART
IrDA port	IrDA UART
Interval timer	Interval timer
System ID	System ID Peripheral
Audio/video configuration port	Audio and Video Config
Audio port	Audio
Video port	Pixel Buffer DMA Controller
Video In port	DMA Controller

Table 4. Qsys cores used in the DE1-SoC Computer.

6 Making the System the Default Configuration

The *DE1-SoC Computer with Nios II* can be loaded into the nonvolatile FPGA configuration memory on the DE1-SoC board, so that it becomes the default system whenever the board is powered on. Instructions for configuring the DE1-SoC board in this manner can be found in the tutorial *Introduction to the Quartus Prime Software*, which is available as part of the `Digital Logic Hardware Design` tutorials in the FPGAcademy.org website.

7 Memory Layout

Table 5 summarizes the memory map used in the DE1-SoC Computer.

Base Address	End Address	I/O Peripheral
0x00000000	0x03FFFFFF	SDRAM
0x08000000	0x0803FFFF	FPGA On-chip Memory
0x09000000	0x09001FFF	FPGA On-chip Memory Character Buffer
0x40000000	0x7FFFFFFF	DDR3 Memory
0xFF200000	0xFF20000F	Red LEDs
0xFF200020	0xFF20002F	7-segment HEX3−HEX0 Displays
0xFF200030	0xFF20003F	7-segment HEX5−HEX4 Displays
0xFF200040	0xFF20004F	Slider Switches
0xFF200050	0xFF20005F	Pushbutton KEYs
0xFF200060	0xFF20006F	JP1 Expansion
0xFF200070	0xFF20007F	JP2 Expansion
0xFF200100	0xFF200107	PS/2
0xFF200108	0xFF20010F	PS/2 Dual
0xFF201000	0xFF201007	JTAG UART
0xFF201020	0xFF201027	Infrared (IrDA)
0xFF202000	0xFF20201F	Interval Timer
0xFF202020	0xFF20202F	Second Interval Timer
0xFF203000	0xFF20301F	Audio/video Configuration
0xFF203020	0xFF20302F	Pixel Buffer Control
0xFF203030	0xFF203037	Character Buffer Control
0xFF203040	0xFF20304F	Audio
0xFF203060	0xFF203070	Video-in
0xFF204000	0xFF20401F	ADC
0xFFC04000	0xFFC040FC	HPS I2C0

Table 5. Memory layout used in the DE1-SoC Computer.

8 Monitor Program Integration

As we mentioned earlier, the *DE1-SoC Computer with Nios II* system, and the sample programs described in this document, are made available as part of the Monitor Program. Figures 22 to 25 show a series of windows that are used in the Monitor Program to create a new project. In the first screen, shown in Figure 22, the user specifies a file system folder where the project will be stored, gives the project a name, and specifies the type of processor that is being used. Pressing Next opens the window in Figure 23. Here, the user can select the *DE1-SoC Computer with Nios II* as a pre-designed system. The Monitor Program then fills in the relevant information in the *System details* box, which includes the appropriate system info and FPGA configuration files, and preloader. The first of these files specifies to the Monitor Program information about the components that are available in the *DE1-SoC Computer with Nios II*, such as the type of processor and memory components, and the address map. The second file is an FPGA programming bitstream for the *DE1-SoC Computer with Nios II*, which can downloaded by the Monitor Program into the DE1-SoC board. Any system which contains a Hard Processor System (HPS) component must also specify the preloader to be run immediately following the circuit being downloaded. This preloader is used to configure the components within the HPS with the setting required for the specific board.

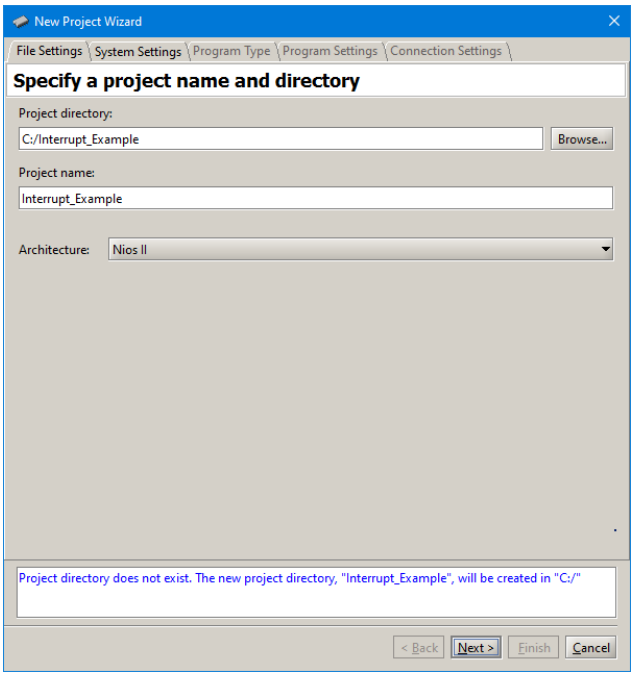


Figure 22. Specifying the project folder and project name.

Pressing Next again opens the window in Figure 24. Here the user selects the type of program that will be used, such as Assembly language, or C. Then, the check box shown in the figure can be used to display the list of sample programs for the *DE1-SoC Computer with Nios II* that are described in this document. When a sample program is selected in this list, its source files, and other settings, can be copied into the project folder in subsequent screens of the Monitor Program.

Figure 25 gives the final screen that is used to create a new project in the Monitor Program. This screen shows the default addresses of compiler and linker sections that will be used for the assembly language or C program associated with the Monitor Program project. In the figure, the drop-down menu called *Linker Section Presets* has been set to Exceptions. With this setting the Monitor Program uses specific compiler and linker sections for the selected processor. For the Nios IIprocessor, these sections are for reset and trap handler, and another section for the main program, called *.text*. For the A9 processor, it has a section for the exception table, called *.vectors*, and another section for the main program, called *.text*. It also shows the initial value used to set the main stack pointer for C programs, which is the starting address of the *.stack* section.

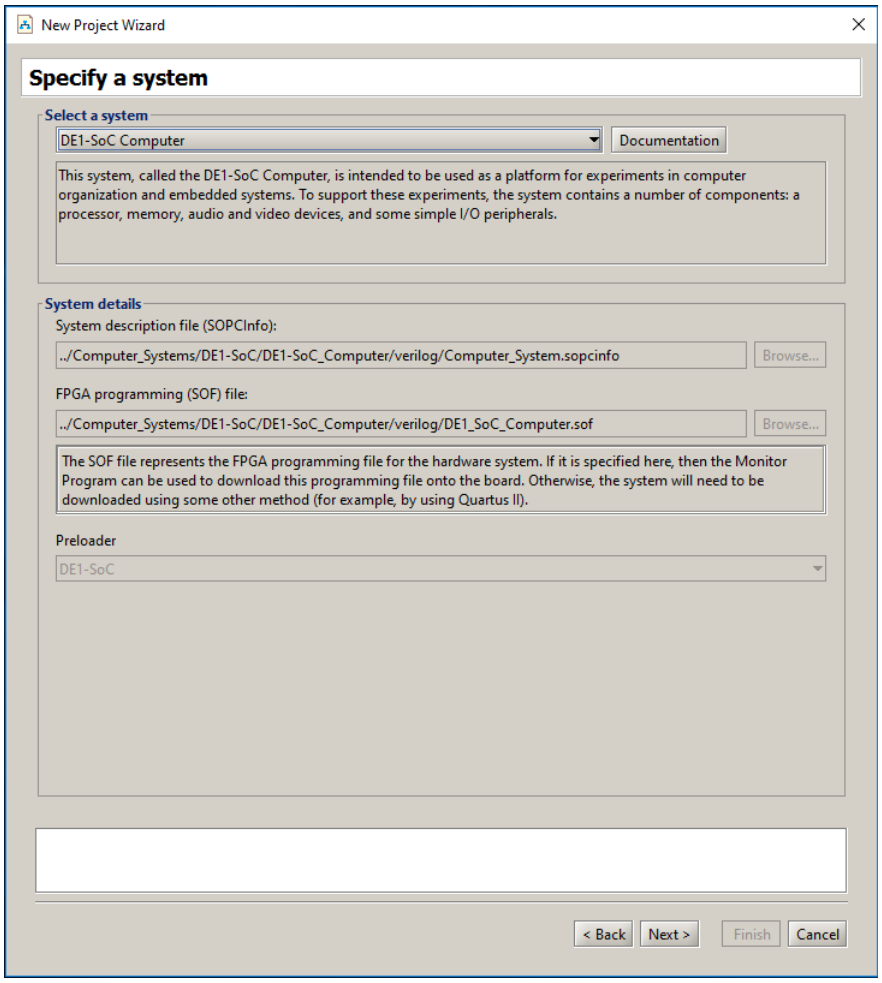


Figure 23. Specifying the DE1-SoC Computer with Nios II.

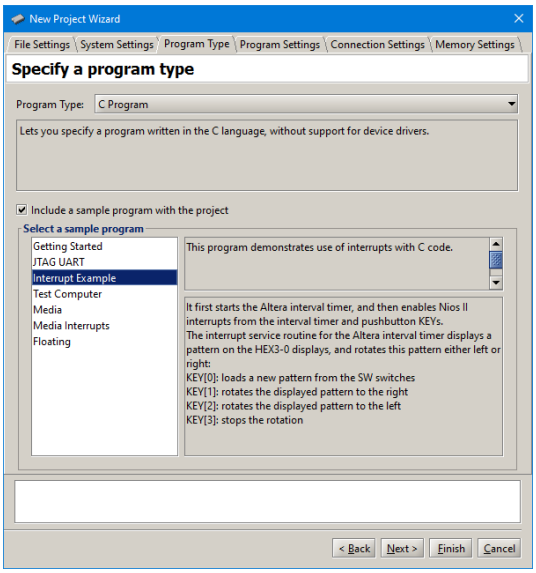


Figure 24. Selecting sample programs.

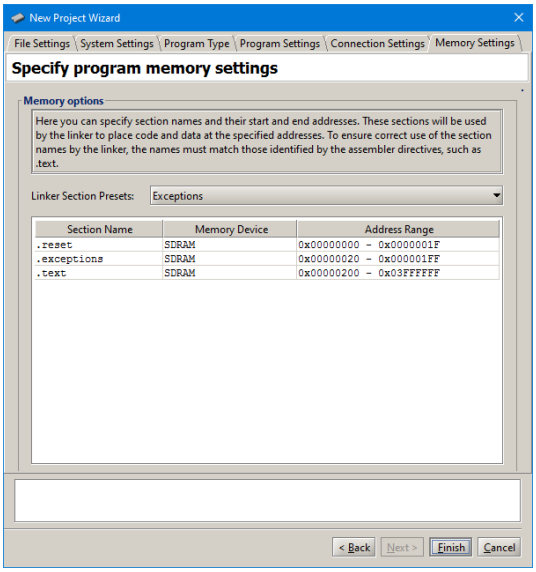


Figure 25. Setting offsets for *.text* and *.data*.

9 Appendix

This section contains all of the source code files mentioned in the document.

9.1 Parallel Ports

```
.include    "address_map_nios2.s"

/*******************************************************************************
 * This program demonstrates use of parallel ports
 *
 * It performs the following:
 *  1. displays a rotating pattern on the LEDs
 *  2. if any KEY is pressed, the SW switches are used as the rotating pattern
 *******************************************************************************/

.text                           # executable code follows
.global     _start
_start:

/* initialize base addresses of parallel ports */
        movia   r15, SW_BASE        # SW slider switch base address
        movia   r16, LED_BASE       # LED base address
        movia   r17, KEY_BASE       # pushbutton KEY base address
        movia   r18, LED_bits
        ldwio   r6, 0(r18)          # load pattern for LED lights

DO_DISPLAY:
        ldwio   r4, 0(r15)          # load slider switches

        ldwio   r5, 0(r17)          # load pushbuttons
        beq     r5, r0, NO_BUTTON
        mov     r6, r4              # copy SW switch values onto LEDs
        roli    r4, r4, 8           # the SW values are copied into the upper three
                                    # bytes of the pattern register
        or      r6, r6, r4          # needed to make pattern consistent as all
                                    # 32-bits of a register are rotated
        roli    r4, r4, 8           # but only the lowest 8-bits are displayed on
                                    # LEDs
        or      r6, r6, r4
        roli    r4, r4, 8
        or      r6, r6, r4
WAIT:
        ldwio   r5, 0(r17)          # load pushbuttons
        bne     r5, r0, WAIT        # wait for button release

NO_BUTTON:
        stwio   r6, 0(r16)          # store to LED
        roli    r6, r6, 1           # rotate the displayed pattern
```

```
        movia   r7, 1500000         # delay counter
DELAY:
        subi    r7, r7, 1
        bne     r7, r0, DELAY

        br      DO_DISPLAY

/*****************************************************************************/
.data                           # data follows

LED_bits:
.word       0x0F0F0F0F

.end
```

Listing 1. An example of Nios II assembly language code that uses parallel ports.

```c
#include "address_map_nios2.h"
/* This program demonstrates use of parallel ports in the Computer System
 *
 * It performs the following:
 *  1. displays a rotating pattern on the LEDs
 *  2. if a KEY is pressed, uses the SW switches as the pattern
*/
int main(void) {
    /* Declare volatile pointers to I/O registers (volatile means that IO load
     * and store instructions will be used to access these pointer locations,
     * instead of regular memory loads and stores)
    */
    volatile int * LED_ptr       = (int *)LED_BASE; // LED address
    volatile int * SW_switch_ptr = (int *)SW_BASE;  // SW slider switch address
    volatile int * KEY_ptr       = (int *)KEY_BASE; // pushbutton KEY address

    int LED_bits = 0x0F0F0F0F; // pattern for LED lights
    int SW_value, KEY_value;
    volatile int
        delay_count; // volatile so the C compiler doesn't remove the loop

    while (1) {
        SW_value = *(SW_switch_ptr); // read the SW slider (DIP) switch values

        KEY_value = *(KEY_ptr); // read the pushbutton KEY values
        if (KEY_value != 0)     // check if any KEY was pressed
        {
            /* set pattern using SW values */
            LED_bits = SW_value | (SW_value << 8) | (SW_value << 16) |
                       (SW_value << 24);
            while (*KEY_ptr)
                ; // wait for pushbutton KEY release
        }
        *(LED_ptr) = LED_bits; // light up the LEDs

        /* rotate the pattern shown on the LEDs */
        if (LED_bits & 0x80000000)
            LED_bits = (LED_bits << 1) | 1;
        else
            LED_bits = LED_bits << 1;

        for (delay_count = 350000; delay_count != 0; --delay_count)
            ; // delay loop
    }
}
```

Listing 2. An example of C code that uses parallel ports.

9.2 Include Files

```
/* Memory */
    .equ    DDR_BASE,               0x40000000
    .equ    DDR_END,                0x7FFFFFFF
    .equ    A9_ONCHIP_BASE,         0xFFFF0000
    .equ    A9_ONCHIP_END,          0xFFFFFFFF
    .equ    SDRAM_BASE,             0x00000000
    .equ    SDRAM_END,              0x03FFFFFF
    .equ    FPGA_PIXEL_BUF_BASE,    0x08000000
    .equ    FPGA_PIXEL_BUF_END,     0x0803FFFF
    .equ    FPGA_CHAR_BASE,         0x09000000
    .equ    FPGA_CHAR_END,          0x09001FFF

/* Cyclone V FPGA devices */
    .equ    LED_BASE,               0xFF200000
    .equ    LEDR_BASE,              0xFF200000
    .equ    HEX3_HEX0_BASE,         0xFF200020
    .equ    HEX5_HEX4_BASE,         0xFF200030
    .equ    SW_BASE,                0xFF200040
    .equ    KEY_BASE,               0xFF200050
    .equ    JP1_BASE,               0xFF200060
    .equ    JP2_BASE,               0xFF200070
    .equ    PS2_BASE,               0xFF200100
    .equ    PS2_DUAL_BASE,          0xFF200108
    .equ    JTAG_UART_BASE,         0xFF201000
    .equ    IrDA_BASE,              0xFF201020
    .equ    TIMER_BASE,             0xFF202000
    .equ    TIMER_2_BASE,           0xFF202020
    .equ    AV_CONFIG_BASE,         0xFF203000
    .equ    PIXEL_BUF_CTRL_BASE,    0xFF203020
    .equ    CHAR_BUF_CTRL_BASE,     0xFF203030
    .equ    AUDIO_BASE,             0xFF203040
    .equ    VIDEO_IN_BASE,          0xFF203060
    .equ    EDGE_DETECT_CTRL_BASE,  0xFF203070
    .equ    ADC_BASE,               0xFF204000

/* Cyclone V HPS devices */
    .equ    HPS_GPIO1_BASE,         0xFF709000
    .equ    I2C0_BASE,              0xFFC04000
    .equ    I2C1_BASE,              0xFFC05000
    .equ    I2C2_BASE,              0xFFC06000
    .equ    I2C3_BASE,              0xFFC07000
    .equ    HPS_TIMER0_BASE,        0xFFC08000
    .equ    HPS_TIMER1_BASE,        0xFFC09000
    .equ    HPS_TIMER2_BASE,        0xFFD00000
    .equ    HPS_TIMER3_BASE,        0xFFD01000
    .equ    FPGA_BRIDGE,            0xFFD0501C
```

Listing 3. The *address_map_nios2.s* include file.

```
/* Memory */
#define DDR_BASE                0x40000000
#define DDR_END                 0x7FFFFFFF
#define A9_ONCHIP_BASE          0xFFFF0000
#define A9_ONCHIP_END           0xFFFFFFFF
#define SDRAM_BASE              0x00000000
#define SDRAM_END               0x03FFFFFF
#define FPGA_PIXEL_BUF_BASE     0x08000000
#define FPGA_PIXEL_BUF_END      0x0803FFFF
#define FPGA_CHAR_BASE          0x09000000
#define FPGA_CHAR_END           0x09001FFF

/* Cyclone V FPGA devices */
#define LED_BASE                0xFF200000
#define LEDR_BASE               0xFF200000
#define HEX3_HEX0_BASE          0xFF200020
#define HEX5_HEX4_BASE          0xFF200030
#define SW_BASE                 0xFF200040
#define KEY_BASE                0xFF200050
#define JP1_BASE                0xFF200060
#define JP2_BASE                0xFF200070
#define PS2_BASE                0xFF200100
#define PS2_DUAL_BASE           0xFF200108
#define JTAG_UART_BASE          0xFF201000
#define IrDA_BASE               0xFF201020
#define TIMER_BASE              0xFF202000
#define TIMER_2_BASE            0xFF202020
#define AV_CONFIG_BASE          0xFF203000
#define RGB_RESAMPLER_BASE      0xFF203010
#define PIXEL_BUF_CTRL_BASE     0xFF203020
#define CHAR_BUF_CTRL_BASE      0xFF203030
#define AUDIO_BASE              0xFF203040
#define VIDEO_IN_BASE           0xFF203060
#define EDGE_DETECT_CTRL_BASE   0xFF203070
#define ADC_BASE                0xFF204000

/* Cyclone V HPS devices */
#define HPS_GPIO1_BASE          0xFF709000
#define I2C0_BASE               0xFFC04000
#define I2C1_BASE               0xFFC05000
#define I2C2_BASE               0xFFC06000
#define I2C3_BASE               0xFFC07000
#define HPS_TIMER0_BASE         0xFFC08000
#define HPS_TIMER1_BASE         0xFFC09000
#define HPS_TIMER2_BASE         0xFFD00000
#define HPS_TIMER3_BASE         0xFFD01000
#define FPGA_BRIDGE             0xFFD0501C

#endif
```

Listing 4. The *address_map_nios2.h* include file.

9.3 JTAG* UART

```
.include    "address_map_nios2.s"

/*********************************************************************
 * This program demonstrates use of the JTAG UART port
 *
 * It performs the following:
 *  1. sends a text string to the JTAG UART
 *  2. reads character data from the JTAG UART
 *  3. echos the character data back to the JTAG UART
 *********************************************************************/

.text        # executable code follows
.global     _start
_start:
/* set up stack pointer */
        movia   sp, SDRAM_END - 3   # starts from largest memory address

        movia   r6, JTAG_UART_BASE  # JTAG UART base address

/* print a text string */
        movia   r8, TEXT_STRING
LOOP:
        ldb     r5, 0(r8)
        beq     r5, zero, GET_JTAG  # string is null-terminated
        call    PUT_JTAG
        addi    r8, r8, 1
        br      LOOP

/* read and echo characters */
GET_JTAG:
        ldwio   r4, 0(r6)           # read the JTAG UART data register
        andi    r8, r4, 0x8000      # check if there is new data
        beq     r8, r0, GET_JTAG    # if no data, wait
        andi    r5, r4, 0x00ff      # the data is in the least significant byte
```

Listing 5. An example of assembly language code that uses the JTAG UART (Part *a*).

```
        call    PUT_JTAG            # echo character
        br      GET_JTAG

/*******************************************************************************
 * Subroutine to send a character to the JTAG UART
 *      r5  = character to send
 *      r6  = JTAG UART base address
 *******************************************************************************/
.global     PUT_JTAG
PUT_JTAG:
/* save any modified registers */
        subi    sp, sp, 4           # reserve space on the stack
        stw     r4, 0(sp)           # save register

        ldwio   r4, 4(r6)           # read the JTAG UART control register
        andhi   r4, r4, 0xffff      # check for write space
        beq     r4, r0, END_PUT     # if no space, ignore the character
        stwio   r5, 0(r6)           # send the character

END_PUT:
/* restore registers */
        ldw     r4, 0(sp)
        addi    sp, sp, 4

        ret

/*******************************************************************************/
.data

TEXT_STRING:
.asciz      "\nJTAG UART example code\n> "

.end
```

Listing 5. An example of assembly language code that uses the JTAG UART (Part *b*).

```c
#include "JTAG_UART.h"
#include "address_map_nios2.h"

/********************************************************************************
 * Subroutine to send a character to the JTAG UART
 ********************************************************************************/
void put_jtag(volatile int * JTAG_UART_ptr, char c)
{
    int control;
    control = *(JTAG_UART_ptr + 1); // read the JTAG_UART control register
    if (control & 0xFFFF0000)       // if space, echo character, else ignore
        *(JTAG_UART_ptr) = c;
}

/********************************************************************************
 * Subroutine to read a character from the JTAG UART
 * Returns \0 if no character, otherwise returns the character
 ********************************************************************************/
char get_jtag(volatile int * JTAG_UART_ptr)
{
    int data;
    data = *(JTAG_UART_ptr); // read the JTAG_UART data register
    if (data & 0x00008000)   // check RVALID to see if there is new data
        return ((char)data & 0xFF);
    else
        return ('\0');
}
```

Listing 6. An example of C code that uses the JTAG UART (Part a).

```c
#include "JTAG_UART.h"
#include "address_map_nios2.h"

/********************************************************************************
 * This program demonstrates use of the JTAG UART port
 *
 * It performs the following:
 *  1. sends a text string to the JTAG UART
 *  2. reads character data from the JTAG UART
 *  3. echos the character data back to the JTAG UART
 ********************************************************************************/
int main(void)
{
    /* Declare volatile pointers to I/O registers (volatile means that IO load
       and store instructions will be used to access these pointer locations,
       instead of regular memory loads and stores) */
    volatile int * JTAG_UART_ptr = (int *)JTAG_UART_BASE; // JTAG UART address

    char  text_string[] = "\nJTAG UART example code\n> \0";
    char *str, c;

    /* print a text string */
    for (str = text_string; *str != 0; ++str)
        put_jtag(JTAG_UART_ptr, *str);

    /* read and echo characters */
    while (1)
    {
        c = get_jtag(JTAG_UART_ptr);
        if (c != '\0')
            put_jtag(JTAG_UART_ptr, c);
    }
}
```

Listing 6. An example of C code that uses the JTAG UART (Part *b*).

9.4 Interrupts

```
.include    "address_map_nios2.s"
.include    "globals.s"

/*******************************************************************************
 * This program demonstrates use of interrupts. It
 * first starts an interval timer with 50 msec timeouts, and then enables
 * Nios II interrupts from the interval timer and pushbutton KEYs
 *
 * The interrupt service routine for the interval timer displays a pattern
 * on the LEDs, and shifts this pattern either left or right:
 *      KEY[0]: loads a new pattern from the SW switches
 *      KEY[1]: toggles the shift direction the displayed pattern
 *******************************************************************************/

.text        # executable code follows
.global     _start
_start:
/* set up the stack */
        movia   sp, SDRAM_END - 3   # stack starts from largest memory
                                    # address

        movia   r16, TIMER_BASE     # interval timer base address
/* set the interval timer period for scrolling the LED lights */
        movia   r12, 5000000        # 1/(100 MHz) x (5 x 10^6) = 50 msec
        sthio   r12, 8(r16)         # store the low half word of counter
                                    # start value
        srli    r12, r12, 16
        sthio   r12, 0xC(r16)       # high half word of counter start value

/* start interval timer, enable its interrupts */
        movi    r15, 0b0111         # START = 1, CONT = 1, ITO = 1
        sthio   r15, 4(r16)

/* write to the pushbutton port interrupt mask register */
        movia   r15, KEY_BASE       # pushbutton key base address
        movi    r7, 0b11            # set interrupt mask bits
        stwio   r7, 8(r15)          # interrupt mask register is (base + 8)

/* enable Nios II processor interrupts */
        movia   r7, 0x00000001      # get interrupt mask bit for interval
                                    # timer
        movia   r8, 0x00000002      # get interrupt mask bit for pushbuttons
        or      r7, r7, r8
        wrctl   ienable, r7         # enable interrupts for the given mask
                                    # bits
        movi    r7, 1
        wrctl   status, r7          # turn on Nios II interrupt processing

IDLE:
```

```
        br      IDLE                # main program simply idles

.data
/*****************************************************************************
 * The global variables used by the interrupt service routines for the interval
 * timer and the pushbutton keys are declared below
 *****************************************************************************/
.global     PATTERN
PATTERN:
.word       0x0F0F0F0F # pattern to show on the LED lights
.global     SHIFT_DIR
SHIFT_DIR:
.word       RIGHT # pattern shifting direction

.end
```

Listing 7. An example of assembly language code that uses interrupts.

```
/********************************************************************************
 * RESET SECTION
 * Note: "ax" is REQUIRED to designate the section as allocatable and executable.
 * Also, the Debug Client automatically places the ".reset" section at the reset
 * location specified in the CPU settings in SOPC Builder.
 ********************************************************************************/
.section    .reset, "ax"

        movia   r2, _start
        jmp     r2                      # branch to main program

/********************************************************************************
 * EXCEPTIONS SECTION
 * Note: "ax" is REQUIRED to designate the section as allocatable and executable.
 * Also, the Monitor Program automatically places the ".exceptions" section at
 * the exception location specified in the CPU settings in SOPC Builder.
 ********************************************************************************/
.section    .exceptions, "ax"
.global     EXCEPTION_HANDLER

EXCEPTION_HANDLER:
        subi    sp, sp, 16              # make room on the stack
        stw     et, 0(sp)

        rdctl   et, ctl4
        beq     et, r0, SKIP_EA_DEC     # interrupt is not external

        subi    ea, ea, 4               # must decrement ea by one instruction
                                        # for external interrupts, so that the
                                        # interrupted instruction will be run
SKIP_EA_DEC:
        stw     ea, 4(sp)               # save all used registers on the Stack
        stw     ra, 8(sp)               # needed if call inst is used
        stw     r22, 12(sp)

        rdctl   et, ctl4
        bne     et, r0, CHECK_LEVEL_0   # interrupt is an external interrupt

NOT_EI:                                 # exception must be unimplemented
                                        # instruction or TRAP instruction. This
                                        # code does not handle those cases
        br      END_ISR

CHECK_LEVEL_0:                          # interval timer is interrupt level 0
        andi    r22, et, 0b1
        beq     r22, r0, CHECK_LEVEL_1

        call    INTERVAL_TIMER_ISR
        br      END_ISR

CHECK_LEVEL_1:                          # pushbutton port is interrupt level 1
```

```
        andi    r22, et, 0b10
        beq     r22, r0, END_ISR        # other interrupt levels are not handled in
                                        # this code

        call    PUSHBUTTON_ISR

END_ISR:
        ldw     et, 0(sp)               # restore all used register to previous
                                        # values
        ldw     ea, 4(sp)
        ldw     ra, 8(sp)               # needed if call inst is used
        ldw     r22, 12(sp)
        addi    sp, sp, 16

.end
```

Listing 8. Reset and exception handler assembly language code.

```
.include    "address_map_nios2.s"
.include    "globals.s"
.extern     PATTERN         # externally defined variables
.extern     SHIFT_DIR
/*******************************************************************************
 * Interval timer - Interrupt Service Routine
 *
 * Shifts a PATTERN being displayed. The shift direction is determined by the
 * external variable SHIFT_DIR.
 *******************************************************************************/
.global     INTERVAL_TIMER_ISR
INTERVAL_TIMER_ISR:
        subi    sp, sp, 40      # reserve space on the stack
        stw     ra, 0(sp)
        stw     r4, 4(sp)
        stw     r5, 8(sp)
        stw     r6, 12(sp)
        stw     r8, 16(sp)
        stw     r10, 20(sp)
        stw     r20, 24(sp)
        stw     r21, 28(sp)
        stw     r22, 32(sp)
        stw     r23, 36(sp)

        movia   r10, TIMER_BASE # interval timer base address
        sthio   r0, 0(r10)      # clear the interrupt

        movia   r20, LED_BASE   # LED base address
        movia   r21, PATTERN    # set up a pointer to the display pattern
        movia   r22, SHIFT_DIR  # set up a pointer to the shift direction variable

        ldw     r6, 0(r21)      # load the pattern
        stwio   r6, 0(r20)      # store to LEDs

CHECK_SHIFT:
        ldw     r5, 0(r22)      # get shift direction
        movi    r8, RIGHT
        bne     r5, r8, SHIFT_L

SHIFT_R:
        movi    r5, 1           # set r5 to the constant value 1
        ror     r6, r6, r5      # rotate the displayed pattern right
        br      STORE_PATTERN

SHIFT_L:
        movi    r5, 1           # set r5 to the constant value 1
        rol     r6, r6, r5      # shift left

STORE_PATTERN:
        stw     r6, 0(r21)      # store display pattern
```

```
END_INTERVAL_TIMER_ISR:
        ldw     ra, 0(sp)       # restore registers
        ldw     r4, 4(sp)
        ldw     r5, 8(sp)
        ldw     r6, 12(sp)
        ldw     r8, 16(sp)
        ldw     r10, 20(sp)
        ldw     r20, 24(sp)
        ldw     r21, 28(sp)
        ldw     r22, 32(sp)
        ldw     r23, 36(sp)
        addi    sp, sp, 40      # release the reserved space on the stack


.end
```

Listing 9. Interrupt service routine for the interval timer.

```
.include    "address_map_nios2.s"
.include    "globals.s"
.extern     PATTERN                         # externally defined variables
.extern     SHIFT_DIR
/*****************************************************************************
 * Pushbutton - Interrupt Service Routine
 *
 * This routine checks which KEY has been pressed and updates the global
 * variables as required.
 *****************************************************************************/
.global     PUSHBUTTON_ISR
PUSHBUTTON_ISR:
        subi    sp, sp, 20                      # reserve space on the stack
        stw     ra, 0(sp)
        stw     r10, 4(sp)
        stw     r11, 8(sp)
        stw     r12, 12(sp)
        stw     r13, 16(sp)

        movia   r10, KEY_BASE                   # base address of pushbutton KEY
                                                # parallel port
        ldwio   r11, 0xC(r10)                   # read edge capture register
        stwio   r11, 0xC(r10)                   # clear the interrupt

CHECK_KEY0:
        andi    r13, r11, 0b0001                # check KEY0
        beq     r13, zero, CHECK_KEY1

        movia   r10, SW_BASE                    # base address of SW slider
                                                # switches parallel port
        ldwio   r12, 0(r10)                     # load a new pattern from the SW
                                                # switches
        movia   r10, PATTERN                    # set up a pointer to the pattern
                                                # variable
        stw     r12, 0(r10)                     # store the new pattern to the
                                                # global variable

CHECK_KEY1:
        andi    r13, r11, 0b0010                # check KEY1
        beq     r13, zero, END_PUSHBUTTON_ISR

        movia   r10, SHIFT_DIR                  # set up a pointer to the shift
                                                # direction variable
        ldw     r12, 0(r10)                     # load the current shift direction
        xori    r12, r12, 1                     # toggle the direction
        stw     r12, 0(r10)                     # store the new shift direction

END_PUSHBUTTON_ISR:
        ldw     ra, 0(sp)                       # Restore all used register to
                                                # previous
        ldw     r10, 4(sp)
```

```
        ldw     r11, 8(sp)
        ldw     r12, 12(sp)
        ldw     r13, 16(sp)
        addi    sp, sp, 20


.end
```

Listing 10. Interrupt service routine for the pushbutton KEYs.

```c
#include "address_map_nios2.h"
#include "globals.h" // defines global values
#include "nios2_ctrl_reg_macros.h"

/* the global variables are written by interrupt service routines; we have to
 * declare
 * these as volatile to avoid the compiler caching their values in registers */
volatile int pattern      = 0x0000000F; // pattern for shifting
volatile int shift_dir    = LEFT;       // direction to shift the pattern
volatile int shift_enable = ENABLE; // enable/disable shifting of the pattern

/********************************************************************************
 * This program demonstrates use of interrupts. It
 * first starts the interval timer with 50 msec timeouts, and then enables
 * Nios II interrupts from the interval timer and pushbutton KEYs
 *
 * The interrupt service routine for the interval timer displays a pattern on
 * the LED lights, and shifts this pattern either left or right. The shifting
 * direction is reversed when KEY[1] is pressed
********************************************************************************/
int main(void) {
    /* Declare volatile pointers to I/O registers (volatile means that IO load
     * and store instructions will be used to access these pointer locations,
     * instead of regular memory loads and stores)
     */
    volatile int * interval_timer_ptr =
        (int *)TIMER_BASE;                    // interal timer base address
    volatile int * KEY_ptr = (int *)KEY_BASE; // pushbutton KEY address

    /* set the interval timer period for scrolling the LED lights */
    int counter                 = 2500000; // 1/(50 MHz) x (2500000) = 50 msec
    *(interval_timer_ptr + 0x2) = (counter & 0xFFFF);
    *(interval_timer_ptr + 0x3) = (counter >> 16) & 0xFFFF;

    /* start interval timer, enable its interrupts */
    *(interval_timer_ptr + 1) = 0x7; // STOP = 0, START = 1, CONT = 1, ITO = 1

    *(KEY_ptr + 2) = 0x3; // enable interrupts for all pushbuttons

    /* set interrupt mask bits for levels 0 (interval timer) and level 1
     * (pushbuttons) */
    NIOS2_WRITE_IENABLE(0x3);

    NIOS2_WRITE_STATUS(1); // enable Nios II interrupts

    while (1)
        ; // main program simply idles
}
```

Listing 11. An example of C code that uses interrupts.

```
#ifndef __NIOS2_CTRL_REG_MACROS__
#define __NIOS2_CTRL_REG_MACROS__

/*****************************************************************************/
/* Macros for accessing the control registers.                               */
/*****************************************************************************/

#define NIOS2_READ_STATUS(dest) \
    do { dest = __builtin_rdctl(0); } while (0)

#define NIOS2_WRITE_STATUS(src) \
    do { __builtin_wrctl(0, src); } while (0)

#define NIOS2_READ_ESTATUS(dest) \
    do { dest = __builtin_rdctl(1); } while (0)

#define NIOS2_READ_BSTATUS(dest) \
    do { dest = __builtin_rdctl(2); } while (0)

#define NIOS2_READ_IENABLE(dest) \
    do { dest = __builtin_rdctl(3); } while (0)

#define NIOS2_WRITE_IENABLE(src) \
    do { __builtin_wrctl(3, src); } while (0)

#define NIOS2_READ_IPENDING(dest) \
    do { dest = __builtin_rdctl(4); } while (0)

#define NIOS2_READ_CPUID(dest) \
    do { dest = __builtin_rdctl(5); } while (0)

#endif
```

Listing 12. Macros for accessing Nios II status and control registers.

```
#include "nios2_ctrl_reg_macros.h"

/* function prototypes */
void main(void);
void interrupt_handler(void);
void interval_timer_ISR(void);
void pushbutton_ISR(void);

/* The assembly language code below handles CPU reset processing */
void the_reset(void) __attribute__((section(".reset")));
void the_reset(void)
/*******************************************************************************
 * Reset code. By giving the code a section attribute with the name ".reset" we
 * allow the linker program to locate this code at the proper reset vector
 * address. This code just calls the main program.
 *******************************************************************************/
{
    asm(".set       noat"); /* Instruct the assembler NOT to use reg at (r1) as
                             * a temp register for performing optimizations */
    asm(".set       nobreak");  /* Suppresses a warning message that says that
                                 * some debuggers corrupt regs bt (r25) and ba
                                 * (r30)
                                 */
    asm("movia      r2, main"); // Call the C language main program
    asm("jmp        r2");
}

/* The assembly language code below handles CPU exception processing. This
 * code should not be modified; instead, the C language code in the function
 * interrupt_handler() can be modified as needed for a given application.
*/
void the_exception(void) __attribute__((section(".exceptions")));
void the_exception(void)
/*******************************************************************************
 * Exceptions code. By giving the code a section attribute with the name
 * ".exceptions" we allow the linker program to locate this code at the proper
 * exceptions vector address.
 * This code calls the interrupt handler and later returns from the exception.
 *******************************************************************************/
{
    asm("subi   sp, sp, 128");
    asm("stw    et, 96(sp)");
    asm("rdctl  et, ctl4");
    asm("beq    et, r0, SKIP_EA_DEC"); // Interrupt is not external
    asm("subi   ea, ea, 4");           /* Must decrement ea by one instruction
                                        * for external interupts, so that the
                                        * interrupted instruction will be run */

    asm("SKIP_EA_DEC:");
    asm("stw    r1,  4(sp)"); // Save all registers
    asm("stw    r2,  8(sp)");
```

```
asm("stw    r3,  12(sp)");
asm("stw    r4,  16(sp)");
asm("stw    r5,  20(sp)");
asm("stw    r6,  24(sp)");
asm("stw    r7,  28(sp)");
asm("stw    r8,  32(sp)");
asm("stw    r9,  36(sp)");
asm("stw    r10, 40(sp)");
asm("stw    r11, 44(sp)");
asm("stw    r12, 48(sp)");
asm("stw    r13, 52(sp)");
asm("stw    r14, 56(sp)");
asm("stw    r15, 60(sp)");
asm("stw    r16, 64(sp)");
asm("stw    r17, 68(sp)");
asm("stw    r18, 72(sp)");
asm("stw    r19, 76(sp)");
asm("stw    r20, 80(sp)");
asm("stw    r21, 84(sp)");
asm("stw    r22, 88(sp)");
asm("stw    r23, 92(sp)");
asm("stw    r25, 100(sp)"); // r25 = bt (skip r24 = et, because it is saved
                            // above)
asm("stw    r26, 104(sp)"); // r26 = gp
// skip r27 because it is sp, and there is no point in saving this
asm("stw    r28, 112(sp)"); // r28 = fp
asm("stw    r29, 116(sp)"); // r29 = ea
asm("stw    r30, 120(sp)"); // r30 = ba
asm("stw    r31, 124(sp)"); // r31 = ra
asm("addi   fp,  sp, 128");

asm("call   interrupt_handler"); // Call the C language interrupt handler

asm("ldw    r1,  4(sp)"); // Restore all registers
asm("ldw    r2,  8(sp)");
asm("ldw    r3,  12(sp)");
asm("ldw    r4,  16(sp)");
asm("ldw    r5,  20(sp)");
asm("ldw    r6,  24(sp)");
asm("ldw    r7,  28(sp)");
asm("ldw    r8,  32(sp)");
asm("ldw    r9,  36(sp)");
asm("ldw    r10, 40(sp)");
asm("ldw    r11, 44(sp)");
asm("ldw    r12, 48(sp)");
asm("ldw    r13, 52(sp)");
asm("ldw    r14, 56(sp)");
asm("ldw    r15, 60(sp)");
asm("ldw    r16, 64(sp)");
asm("ldw    r17, 68(sp)");
asm("ldw    r18, 72(sp)");
```

```
    asm("ldw    r19, 76(sp)");
    asm("ldw    r20, 80(sp)");
    asm("ldw    r21, 84(sp)");
    asm("ldw    r22, 88(sp)");
    asm("ldw    r23, 92(sp)");
    asm("ldw    r24, 96(sp)");
    asm("ldw    r25, 100(sp)"); // r25 = bt
    asm("ldw    r26, 104(sp)"); // r26 = gp
    // skip r27 because it is sp, and we did not save this on the stack
    asm("ldw    r28, 112(sp)"); // r28 = fp
    asm("ldw    r29, 116(sp)"); // r29 = ea
    asm("ldw    r30, 120(sp)"); // r30 = ba
    asm("ldw    r31, 124(sp)"); // r31 = ra

    asm("addi   sp,  sp, 128");

    asm("eret");
}

/*******************************************************************************
 * Interrupt Service Routine
 *  Determines what caused the interrupt and calls the appropriate
 *  subroutine.
 *
 * ipending - Control register 4 which has the pending external interrupts
 *******************************************************************************/
void interrupt_handler(void) {
    int ipending;
    NIOS2_READ_IPENDING(ipending);
    if (ipending & 0x1) // interval timer is interrupt level 0
    {
        interval_timer_ISR();
    }
    if (ipending & 0x2) // pushbuttons are interrupt level 1
    {
        pushbutton_ISR();
    }
    // else, ignore the interrupt
    return;
}
```

Listing 13. Reset and exception handler C code.

```
#include "address_map_nios2.h"
#include "globals.h" // defines global values

extern volatile int pattern, shift_dir, shift_enable;
/*****************************************************************************
 * Interval timer interrupt service routine
 *
 * Shifts a PATTERN being displayed on the LED lights. The shift direction
 * is determined by the external variable key_dir.
 *****************************************************************************/
void interval_timer_ISR() {
    volatile int * interval_timer_ptr = (int *)TIMER_BASE;
    volatile int * LEDG_ptr           = (int *)LED_BASE; // LED address

    *(interval_timer_ptr) = 0; // clear the interrupt

    *(LEDG_ptr) = pattern; // display pattern on LED

    if (shift_enable == DISABLE) // check if shifting is disabled
        return;

    /* rotate the pattern shown on the LEDG lights */
    if (shift_dir == LEFT) // rotate left
        if (pattern & 0x80000000)
            pattern = (pattern << 1) | 1;
        else
            pattern = pattern << 1;
    else // rotate right
        if (pattern & 0x00000001)
        pattern = (pattern >> 1) | 0x80000000;
    else
        pattern = (pattern >> 1) & 0x7FFFFFFF;

    return;
}
```

Listing 14. Interrupt service routine for the interval timer.

```
#include "address_map_nios2.h"
#include "globals.h" // defines global values

extern volatile int pattern, shift_dir, shift_enable;
/*****************************************************************************
 * Pushbutton - Interrupt Service Routine
 *
 * This routine checks which KEY has been pressed and updates the global
 * variables as required.
 ******************************************************************************/
void pushbutton_ISR(void) {
    volatile int * KEY_ptr           = (int *)KEY_BASE;
    volatile int * slider_switch_ptr = (int *)SW_BASE;
    int            press;

    press          = *(KEY_ptr + 3); // read the pushbutton interrupt register
    *(KEY_ptr + 3) = press;          // Clear the interrupt

    if (press & 0x1) // KEY0
        pattern = *slider_switch_ptr;

    if (press & 0x2) // KEY1
        shift_dir = shift_dir ^ 1;

    return;
}
```

Listing 15. Interrupt service routine for the pushbutton KEYs.

9.5 Audio

```
#include "address_map_nios2.h"

/* globals */
#define BUF_SIZE 80000  // about 10 seconds of buffer (@ 8K samples/sec)
#define BUF_THRESHOLD 96 // 75% of 128 word buffer

/* function prototypes */
void check_KEYs(int *, int *, int *);

/********************************************************************************
 * This program performs the following:
 *  1. records audio for 10 seconds when KEY[0] is pressed. LEDR[0] is lit
 *     while recording.
 *  2. plays the recorded audio when KEY[1] is pressed. LEDR[1] is lit while
 *     playing.
 ********************************************************************************/
int main(void) {
    /* Declare volatile pointers to I/O registers (volatile means that IO load
       and store instructions will be used to access these pointer locations,
       instead of regular memory loads and stores) */
    volatile int * red_LED_ptr = (int *)LED_BASE;
    volatile int * audio_ptr   = (int *)AUDIO_BASE;

    /* used for audio record/playback */
    int fifospace;
    int record = 0, play = 0, buffer_index = 0;
    int left_buffer[BUF_SIZE];
    int right_buffer[BUF_SIZE];

    /* read and echo audio data */
    record = 0;
    play   = 0;

    while (1) {
        check_KEYs(&record, &play, &buffer_index);
        if (record) {
            *(red_LED_ptr) = 0x1; // turn on LEDR[0]
            fifospace =
                *(audio_ptr + 1); // read the audio port fifospace register
            if ((fifospace & 0x000000FF) > BUF_THRESHOLD) // check RARC
            {
                // store data until the the audio-in FIFO is empty or the buffer
                // is full
                while ((fifospace & 0x000000FF) && (buffer_index < BUF_SIZE)) {
                    left_buffer[buffer_index]  = *(audio_ptr + 2);
                    right_buffer[buffer_index] = *(audio_ptr + 3);
                    ++buffer_index;

                    if (buffer_index == BUF_SIZE) {
```

```
                        // done recording
                        record          = 0;
                        *(red_LED_ptr) = 0x0; // turn off LEDR
                    }
                    fifospace = *(audio_ptr +
                                  1); // read the audio port fifospace register
                }
            }
        } else if (play) {
            *(red_LED_ptr) = 0x2; // turn on LEDR_1
            fifospace =
                *(audio_ptr + 1); // read the audio port fifospace register
            if ((fifospace & 0x00FF0000) > BUF_THRESHOLD) // check WSRC
            {
                // output data until the buffer is empty or the audio-out FIFO
                // is full
                while ((fifospace & 0x00FF0000) && (buffer_index < BUF_SIZE)) {
                    *(audio_ptr + 2) = left_buffer[buffer_index];
                    *(audio_ptr + 3) = right_buffer[buffer_index];
                    ++buffer_index;

                    if (buffer_index == BUF_SIZE) {
                        // done playback
                        play            = 0;
                        *(red_LED_ptr) = 0x0; // turn off LEDR
                    }
                    fifospace = *(audio_ptr +
                                  1); // read the audio port fifospace register
                }
            }
        }
    }
}

/*****************************************************************************
 * Subroutine to read KEYs
*****************************************************************************/
void check_KEYs(int * KEY0, int * KEY1, int * counter) {
    volatile int * KEY_ptr   = (int *)KEY_BASE;
    volatile int * audio_ptr = (int *)AUDIO_BASE;
    int            KEY_value;

    KEY_value = *(KEY_ptr); // read the pushbutton KEY values
    while (*KEY_ptr)
        ; // wait for pushbutton KEY release

    if (KEY_value == 0x1) // check KEY0
    {
        // reset counter to start recording
        *counter = 0;
        // clear audio-in FIFO
```

```
        *(audio_ptr) = 0x4;
        *(audio_ptr) = 0x0;

        *KEY0 = 1;
    } else if (KEY_value == 0x2) // check KEY1
    {
        // reset counter to start playback
        *counter = 0;
        // clear audio-out FIFO
        *(audio_ptr) = 0x8;
        *(audio_ptr) = 0x0;

        *KEY1 = 1;
    }
}
```

Listing 16. An example of code that uses the audio port.

9.6 Video Out

```
#include "address_map_nios2.h"

/* function prototypes */
void video_text(int, int, char *);
void video_box(int, int, int, int, short);
int  resample_rgb(int, int);
int  get_data_bits(int);

#define STANDARD_X 320
#define STANDARD_Y 240
#define INTEL_BLUE 0x0071C5
/* global variables */
int screen_x;
int screen_y;
int res_offset;
int col_offset;

/*******************************************************************************
 * This program demonstrates use of the video in the computer system.
 * Draws a blue box on the video display, and places a text string inside the
 * box
 ******************************************************************************/
int main(void) {
    volatile int * video_resolution = (int *)(PIXEL_BUF_CTRL_BASE + 0x8);
    screen_x                        = *video_resolution & 0xFFFF;
    screen_y                        = (*video_resolution >> 16) & 0xFFFF;

    volatile int * rgb_status = (int *)(RGB_RESAMPLER_BASE);
    int            db         = get_data_bits(*rgb_status & 0x3F);

    /* check if resolution is smaller than the standard 320 x 240 */
    res_offset = (screen_x == 160) ? 1 : 0;

    /* check if number of data bits is less than the standard 16-bits */
    col_offset = (db == 8) ? 1 : 0;

    /* create a message to be displayed on the video and LCD displays */
    char text_top_row[40]    = "Intel FPGA\0";
    char text_bottom_row[40] = "Computer Systems\0";

    /* update color */
    short background_color = resample_rgb(db, INTEL_BLUE);

    video_text(35, 29, text_top_row);
    video_text(32, 30, text_bottom_row);
    video_box(0, 0, STANDARD_X, STANDARD_Y, 0); // clear the screen
    video_box(31 * 4, 28 * 4, 49 * 4 - 1, 32 * 4 - 1, background_color);
}
```

```
/*****************************************************************************
 * Subroutine to send a string of text to the video monitor
 *****************************************************************************/
void video_text(int x, int y, char * text_ptr) {
    int             offset;
    volatile char * character_buffer =
        (char *)FPGA_CHAR_BASE; // video character buffer

    /* assume that the text string fits on one line */
    offset = (y << 7) + x;
    while (*(text_ptr)) {
        *(character_buffer + offset) =
            *(text_ptr); // write to the character buffer
        ++text_ptr;
        ++offset;
    }
}

/*****************************************************************************
 * Draw a filled rectangle on the video monitor
 * Takes in points assuming 320x240 resolution and adjusts based on differences
 * in resolution and color bits.
 *****************************************************************************/
void video_box(int x1, int y1, int x2, int y2, short pixel_color) {
    int pixel_buf_ptr = *(int *)PIXEL_BUF_CTRL_BASE;
    int pixel_ptr, row, col;
    int x_factor = 0x1 << (res_offset + col_offset);
    int y_factor = 0x1 << (res_offset);
    x1          = x1 / x_factor;
    x2          = x2 / x_factor;
    y1          = y1 / y_factor;
    y2          = y2 / y_factor;

    /* assume that the box coordinates are valid */
    for (row = y1; row <= y2; row++)
        for (col = x1; col <= x2; ++col) {
            pixel_ptr = pixel_buf_ptr +
                        (row << (10 - res_offset - col_offset)) + (col << 1);
            *(short *)pixel_ptr = pixel_color; // set pixel color
        }
}

/*****************************************************************************
 * Resamples 24-bit color to 16-bit or 8-bit color
 *****************************************************************************/
int resample_rgb(int num_bits, int color) {
    if (num_bits == 8) {
        color = (((color >> 16) & 0x000000E0) | ((color >> 11) & 0x0000001C) |
                 ((color >> 6) & 0x00000003));
        color = (color << 8) | color;
    } else if (num_bits == 16) {
```

```
        color = (((color >> 8) & 0x0000F800) | ((color >> 5) & 0x000007E0) |
                 ((color >> 3) & 0x0000001F));
    }
    return color;
}

/*****************************************************************************
 * Finds the number of data bits from the mode
 *****************************************************************************/
int get_data_bits(int mode) {
    switch (mode) {
    case 0x0:
        return 1;
    case 0x7:
        return 8;
    case 0x11:
        return 8;
    case 0x12:
        return 9;
    case 0x14:
        return 16;
    case 0x17:
        return 24;
    case 0x19:
        return 30;
    case 0x31:
        return 8;
    case 0x32:
        return 12;
    case 0x33:
        return 16;
    case 0x37:
        return 32;
    case 0x39:
        return 40;
    }
}
```

Listing 17. An example of code that uses the video-out port.

9.7 PS/2

```
#include "address_map_nios2.h"

/* function prototypes */
void HEX_PS2(char, char, char);

/********************************************************************************
 * This program demonstrates use of the PS/2 port by displaying the last three
 * bytes of data received from the PS/2 port on the HEX displays.
 ********************************************************************************/
int main(void) {
    /* Declare volatile pointers to I/O registers (volatile means that IO load
       and store instructions will be used to access these pointer locations,
       instead of regular memory loads and stores) */
    volatile int * PS2_ptr = (int *)PS2_BASE;

    int  PS2_data, RVALID;
    char byte1 = 0, byte2 = 0, byte3 = 0;

    // PS/2 mouse needs to be reset (must be already plugged in)
    *(PS2_ptr) = 0xFF; // reset

    while (1) {
        PS2_data = *(PS2_ptr);        // read the Data register in the PS/2 port
        RVALID   = PS2_data & 0x8000; // extract the RVALID field
        if (RVALID) {
            /* shift the next data byte into the display */
            byte1 = byte2;
            byte2 = byte3;
            byte3 = PS2_data & 0xFF;
            HEX_PS2(byte1, byte2, byte3);

            if ((byte2 == (char)0xAA) && (byte3 == (char)0x00))
                // mouse inserted; initialize sending of data
                *(PS2_ptr) = 0xF4;
        }
    }
}

/*****************************************************************************************
 * Subroutine to show a string of HEX data on the HEX displays
*****************************************************************************************/
void HEX_PS2(char b1, char b2, char b3) {
    volatile int * HEX3_HEX0_ptr = (int *)HEX3_HEX0_BASE;
    volatile int * HEX5_HEX4_ptr = (int *)HEX5_HEX4_BASE;

    /* SEVEN_SEGMENT_DECODE_TABLE gives the on/off settings for all segments in
     * a single 7-seg display in the DE1-SoC Computer, for the hex digits 0 - F
     */
    unsigned char seven_seg_decode_table[] = {
```

```
        0x3F, 0x06, 0x5B, 0x4F, 0x66, 0x6D, 0x7D, 0x07,
        0x7F, 0x67, 0x77, 0x7C, 0x39, 0x5E, 0x79, 0x71};
    unsigned char hex_segs[] = {0, 0, 0, 0, 0, 0, 0, 0};
    unsigned int  shift_buffer, nibble;
    unsigned char code;
    int           i;

    shift_buffer = (b1 << 16) | (b2 << 8) | b3;
    for (i = 0; i < 6; ++i) {
        nibble = shift_buffer & 0x0000000F; // character is in rightmost nibble
        code   = seven_seg_decode_table[nibble];
        hex_segs[i]  = code;
        shift_buffer = shift_buffer >> 4;
    }
    /* drive the hex displays */
    *(HEX3_HEX0_ptr) = *(int *)(hex_segs);
    *(HEX5_HEX4_ptr) = *(int *)(hex_segs + 4);
}
```

Listing 18. An example of code that uses the PS/2 port.

9.8 Floating Point

```
/*****************************************************************************
 * This program demonstrates use of floating-point numbers
 *
 * It performs the following:
 *      1. reads two FP numbers from the Terminal window
 *      2. performs +, -, *, and / on the numbers
 *      3. prints the results on the Terminal window
 *      Note: Please enable "Echo input" in the terminal window
 *****************************************************************************/
#include <stdio.h>

int flush()
{
    while (getchar() != '\n')
        ;
    return 1;
}

int main(void)
{
    float x, y, add, sub, mult, div;

    while (1)
    {
        printf("Enter FP values X: ");

        while ((scanf("%f", &x) != 1) && flush())
            ; // get valid floating point value and flush the invalid input
        printf("%f\n", x); // echo the typed data to the Terminal window

        printf("Enter FP values Y: ");

        while ((scanf("%f", &y) != 1) && flush())
            ; // get valid floating point value and flush the invalid input
        printf("%f\n", y); // echo the typed data to the Terminal window

        add  = x + y;
        sub  = x - y;
        mult = x * y;
        div  = x / y;
        printf("X + Y = %f\n", add);
        printf("X - Y = %f\n", sub);
        printf("X * Y = %f\n", mult);
        printf("X / Y = %f\n", div);
    }
}
```

Listing 19. An example of code that uses floating-point variables.

Copyright © FPGAcademy.org. All rights reserved. FPGAcademy and the FPGAcademy logo are trademarks of FPGAcademy.org. This document is being provided on an "as-is" basis and as an accommodation and therefore all warranties, representations or guarantees of any kind (whether express, implied or statutory) including, without limitation, warranties of merchantability, non-infringement, or fitness for a particular purpose, are specifically disclaimed.

**Other names and brands may be claimed as the property of others.